



Preliminary Design Review

Ultra-thin Carbon Foils

AST 250/251

Fall 2023

Space Physics at Princeton



AST 250/251 Team (2023-2024)

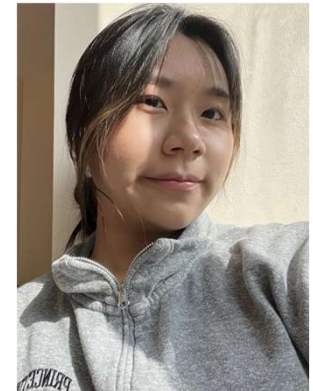




Table of Contents

- | | |
|------------------------------------|---------------------|
| 1. Background | 7. Future Steps |
| 2. Ion Detection + Instrumentation | 8. Spring 2024 |
| 3. Ultra-thin Carbon Foils | 9. Risks + Safety |
| 4. Basic Apparatus Design | 10. Summary |
| 5. Main Apparatus Build Process | 11. Works Cited |
| 6. Challenges | 12. Acknowledgments |



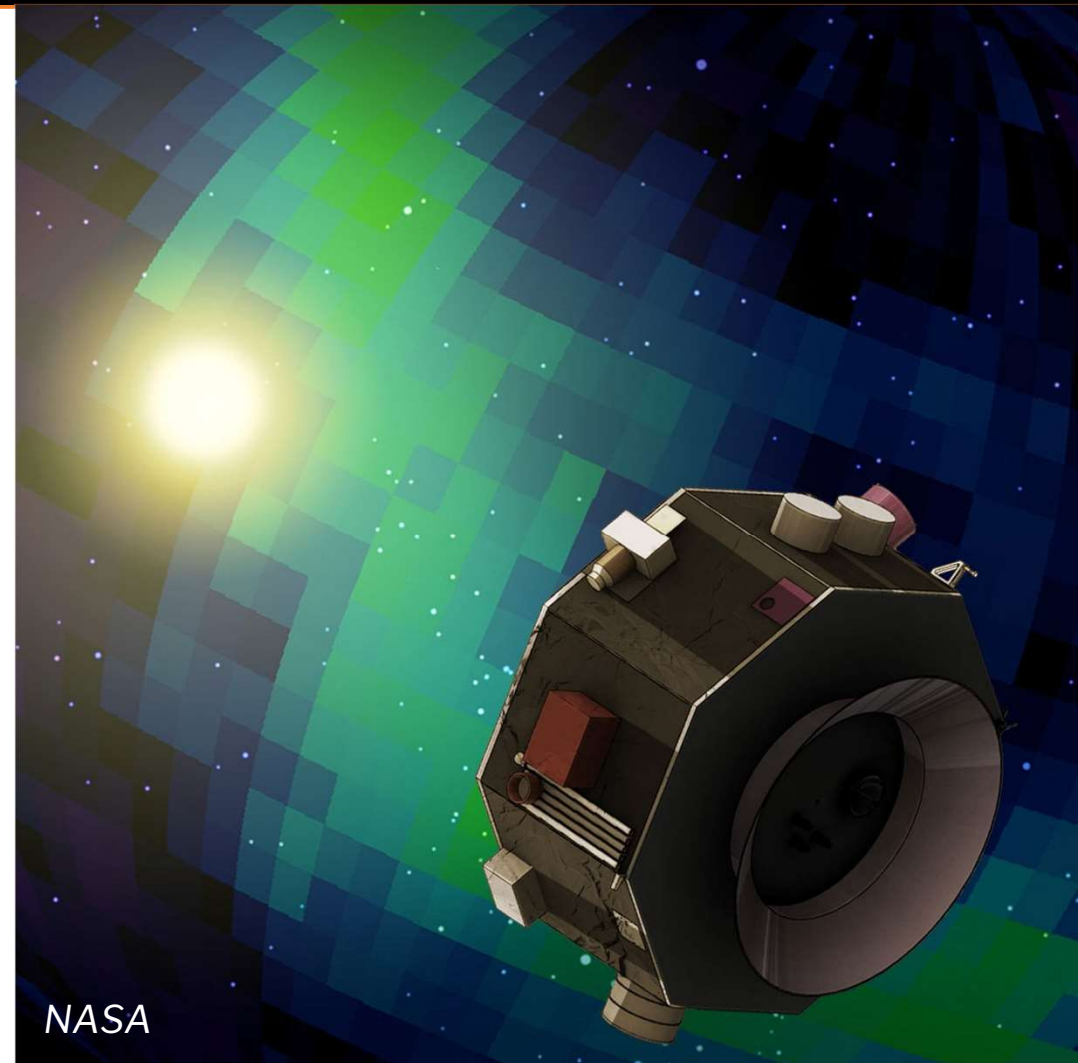
Preliminary Design Review

Background



Interstellar Mapping and Acceleration Probe (IMAP)

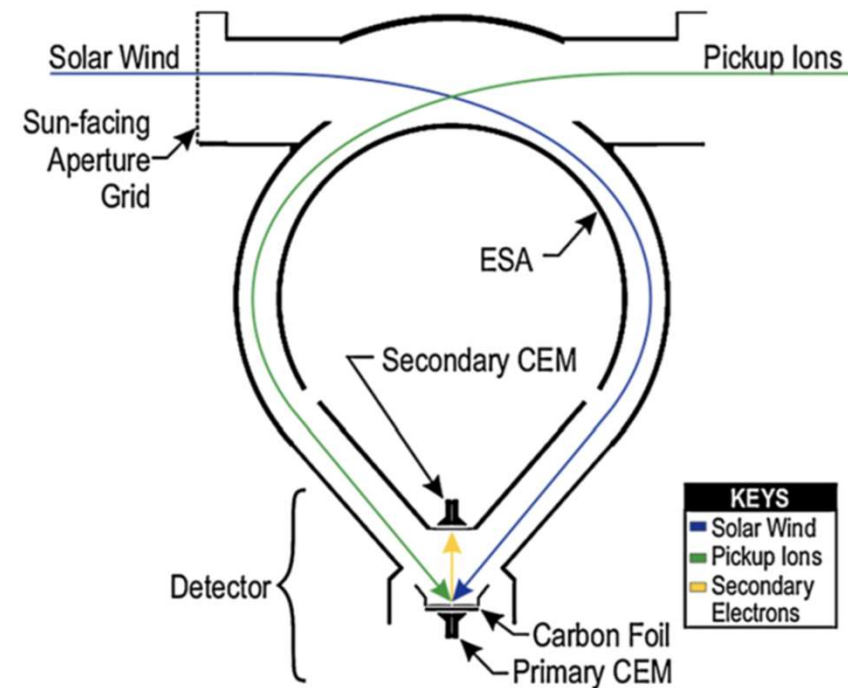
- Investigating the heliosphere:
 - Composition of the Local Interstellar Medium (LISM)
 - Interactions between Solar Wind and the LISM
 - Interactions between the solar magnetic fields and the LISM
 - Particle injection and acceleration





Solar Wind and Pickup Ions (SWAPI)

- Measures elements of solar wind
- Interstellar pickup ions (PUIs)
- Space weather monitoring
 - Solar wind changes over time
- High time resolution measurements of helium PUIs
- Operating at Earth-Sun Lagrange Point 1



Princeton/IMAP

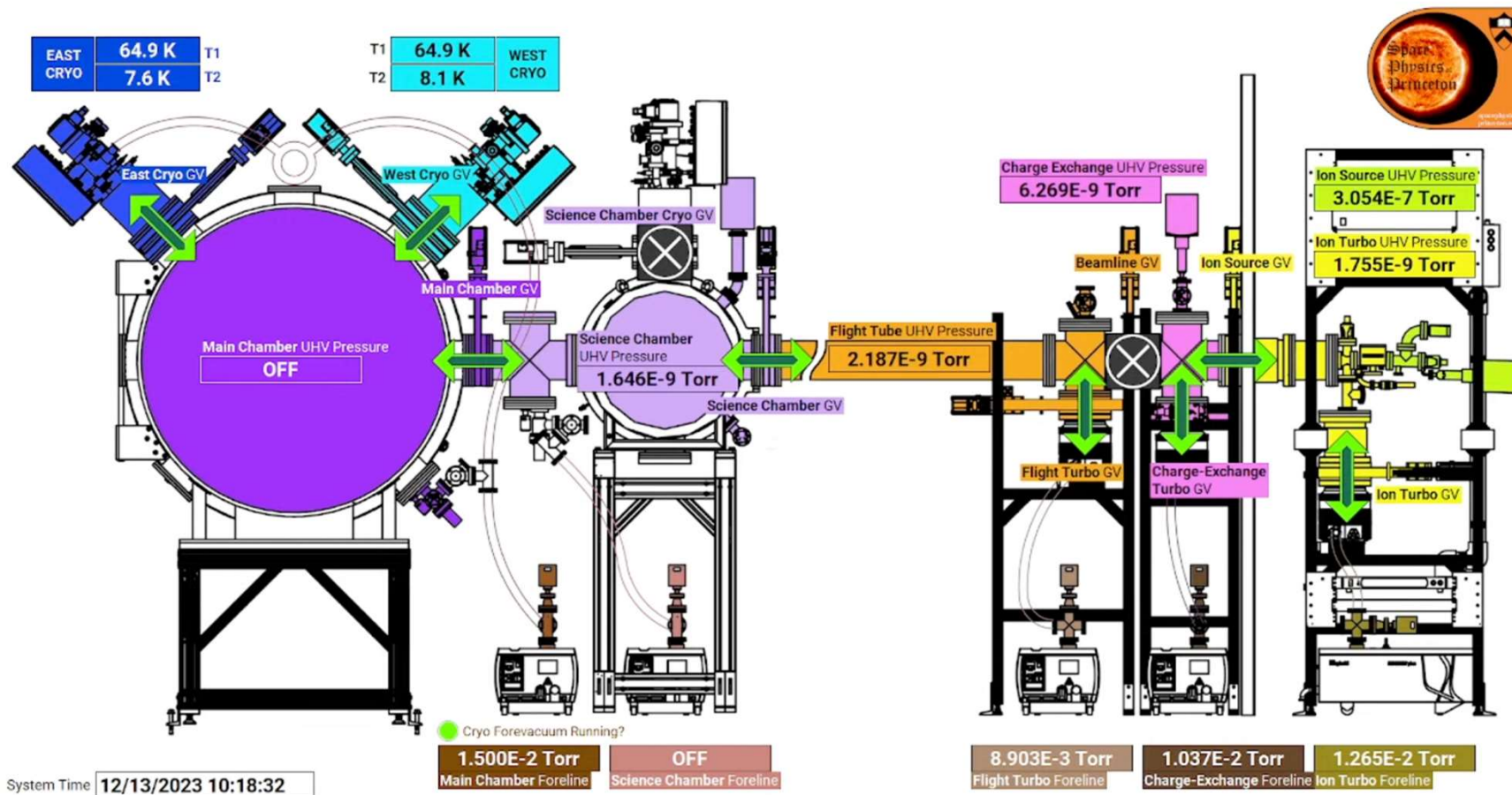


The Princeton Space Physics Lab



Princeton/IMAP

The Princeton Space Physics Lab



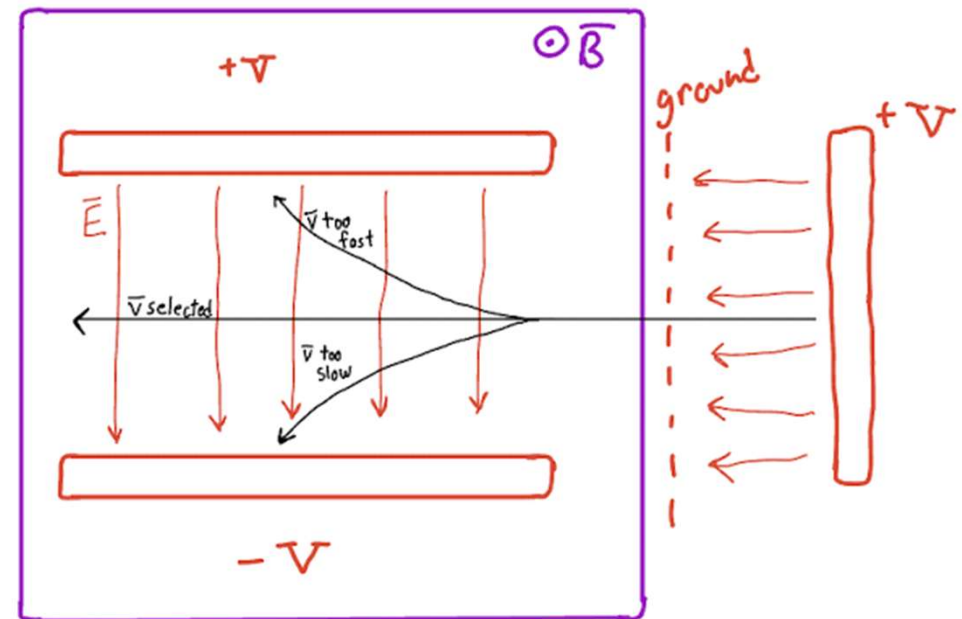


Preliminary Design Review

Ion Detection and Instrumentation

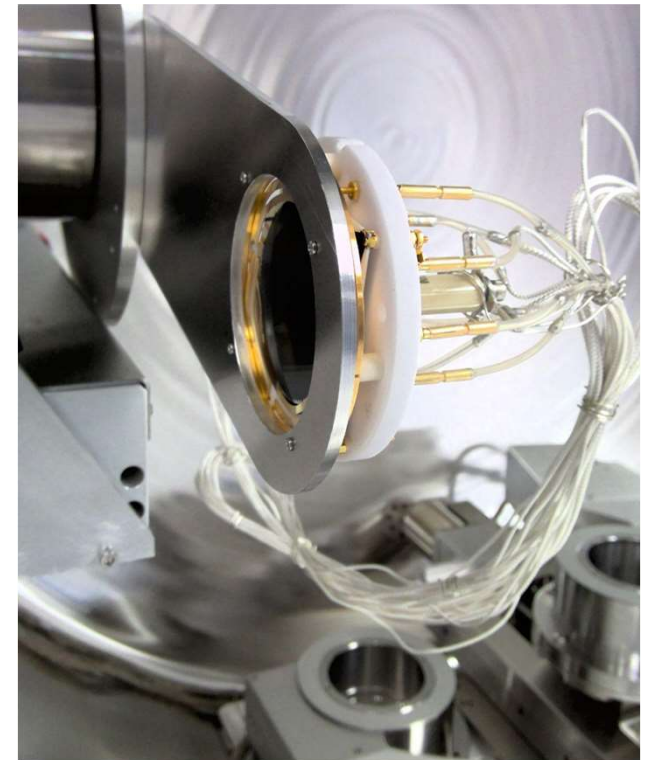
Ion Source

- Ion Source:
 1. Gas: A gas with composition of argon, nitrogen, neon, hydrogen and helium is introduced
 2. Ionization: A filament ionizes the gas
 3. Acceleration: a charged plate creates an **E** field to accelerate ions to a selected energy
 4. Wien Filter: a combined **E** and **B** field are used to filter the desired species of gas



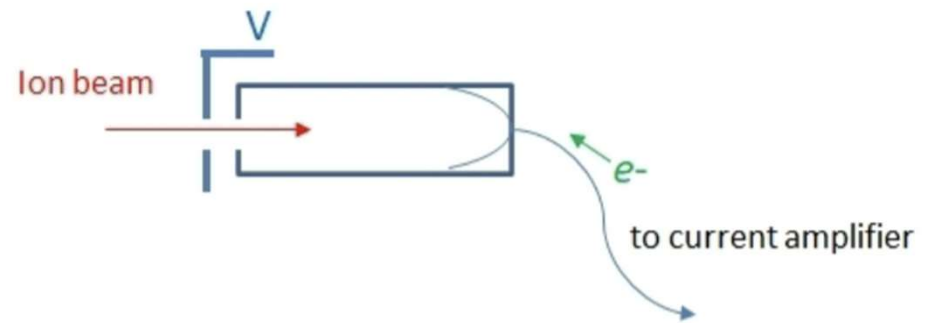
Plasma Beam Diagnostics: MCPs

- MCPs are imaging detectors for photons and charged ions. They amplify the signal of electrons.
- A single incident particle enters a channel and emits an electron from the channel wall — secondary electron cascade.
- Can be used for applications beyond beam monitoring, such as time-of-flight mass spectroscopy.



Plasma Beam Diagnostics: Faraday Cup

- Less sensitive than MCPs.
- When an ion beam strikes its surface, a current is generated, amplified, and measured by the instrument. This current is proportional to the number of ions striking the detector.



Channel Electron Multipliers (CEMs)

- CEM's take in electrons and produce secondary electrons. These secondary electrons can then create more secondary electrons creating an electron avalanche which can then be measured.
- An electric field is usually used to feed electrons into the CEM, as electrons must reach the base of the funnel in order to be detected.





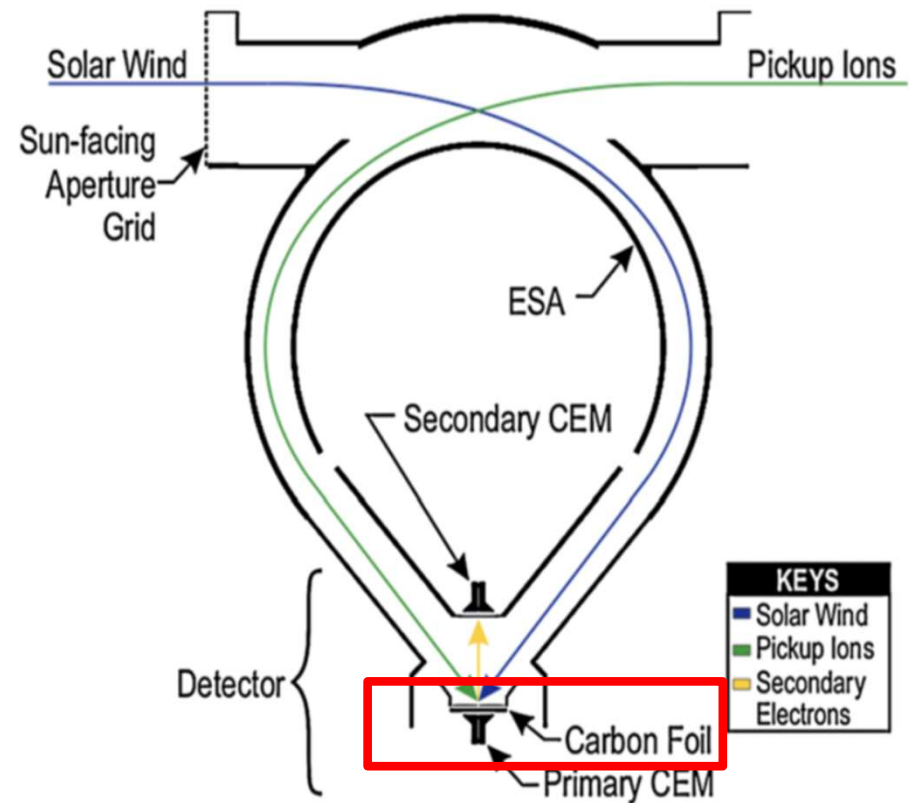
SWAP and SWAPI

- SWAP
 - Detected pickup ions to measure and understand the interactions between Pluto and solar wind.
- SWAPI - based on SWAP but with some modifications:
 - SWAP's Retarding Potential Analyzer and Doors removed
 - Addition of a Solar Wind aperture grid assembly
 - Sunglasses and vanes: prevent scattering
 - Separate entrances for solar wind and pickup ions
 - Voltage increase

Carbon Foils on SWAPI

Carbon foils are the primary detection mechanism in SWAPI:

- 1) Particles enter through the electrostatic analyzer
- 2) Particles hit the carbon foil
- 3) Secondary electrons are emitted from the carbon foil
- 4) CEMs coincident measurements of secondary electrons are registered as a particle detection





Preliminary Design Review

Ultra-thin Carbon Foils



Ultra-thin Carbon Foils

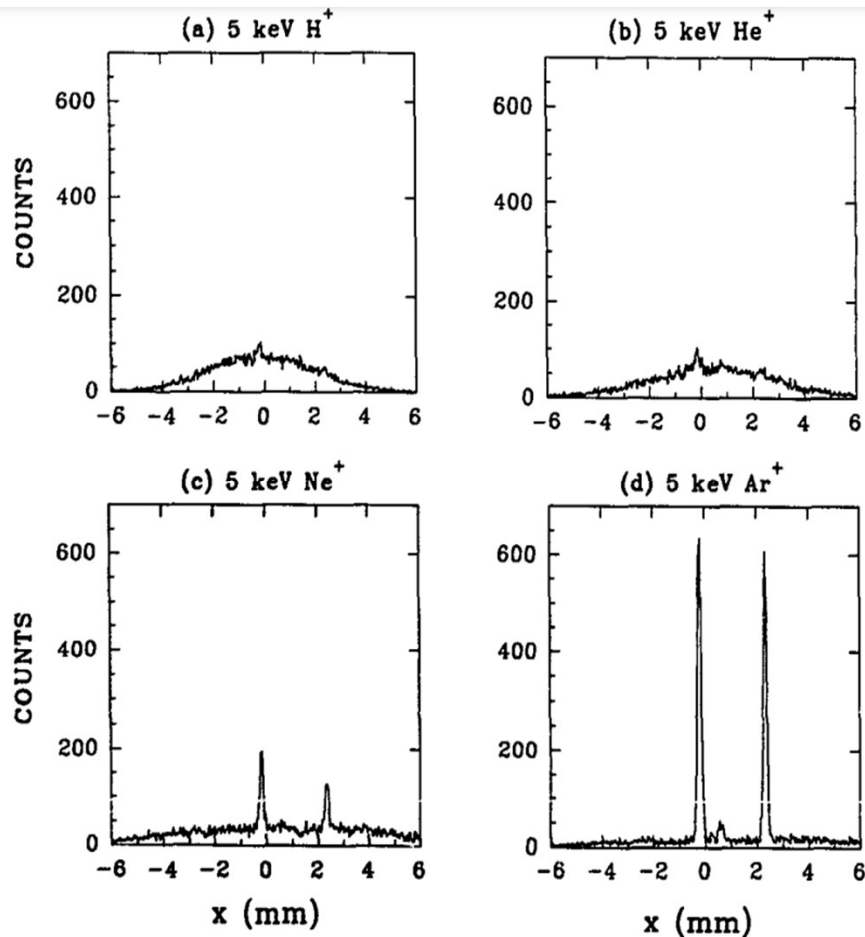
- Carbon foils are made from carbon-12 of the highest purity, and have been used for decades in space instrumentation for ENA detection as well as plasma/plasma mass spectrometry
- Two properties of carbon foils make them ideal for the study of these topics:
 - Secondary electron generation → necessary for ToF measurements
 - Particle charge state modification → makes neutral particle detectable by instruments
- Key features*:
 - Surface density → $\sim 0.5 \mu\text{g cm}^{-2}$
 - Thickness → 2.5 to 17 nm (for this project, thinner the carbon foil, the better)



Manufacturing Process & Installation

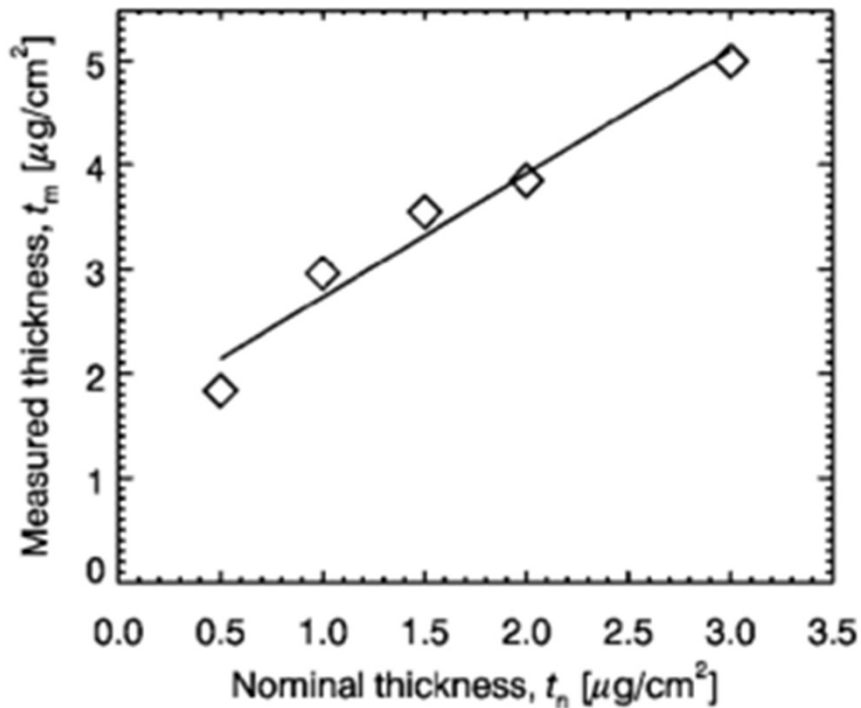
- Carbon foils are produced using the technique of arc evaporation on a carbon cathode. Resulting particles then stick to a glass plate on which a parting agent has been applied
- Foils are then floated in an isopropyl-water mixture, and installed on nickel grids. This procedure makes them surprisingly sturdy, with some foils surviving for up to 25 years in space.
- Because of these physical processes, thickness fluctuations and pinholes will inevitably occur, and yield for usable foils is usually around 50%

Previous Findings: Pinholes and Thin Defects



- Shoot ions through the bulk foil to obtain a two component distribution:
 - Broader scattering angle distribution, which is interesting on its own (see next slide)
 - Deviations from that distribution reflecting surface features such as pinholes and extremely thin defects
- Funsten et al. 1991 found that identifying pinholes is easier for heavier ions
- These 1D images are slices from a 2D image

Previous Findings: Bulk Thickness

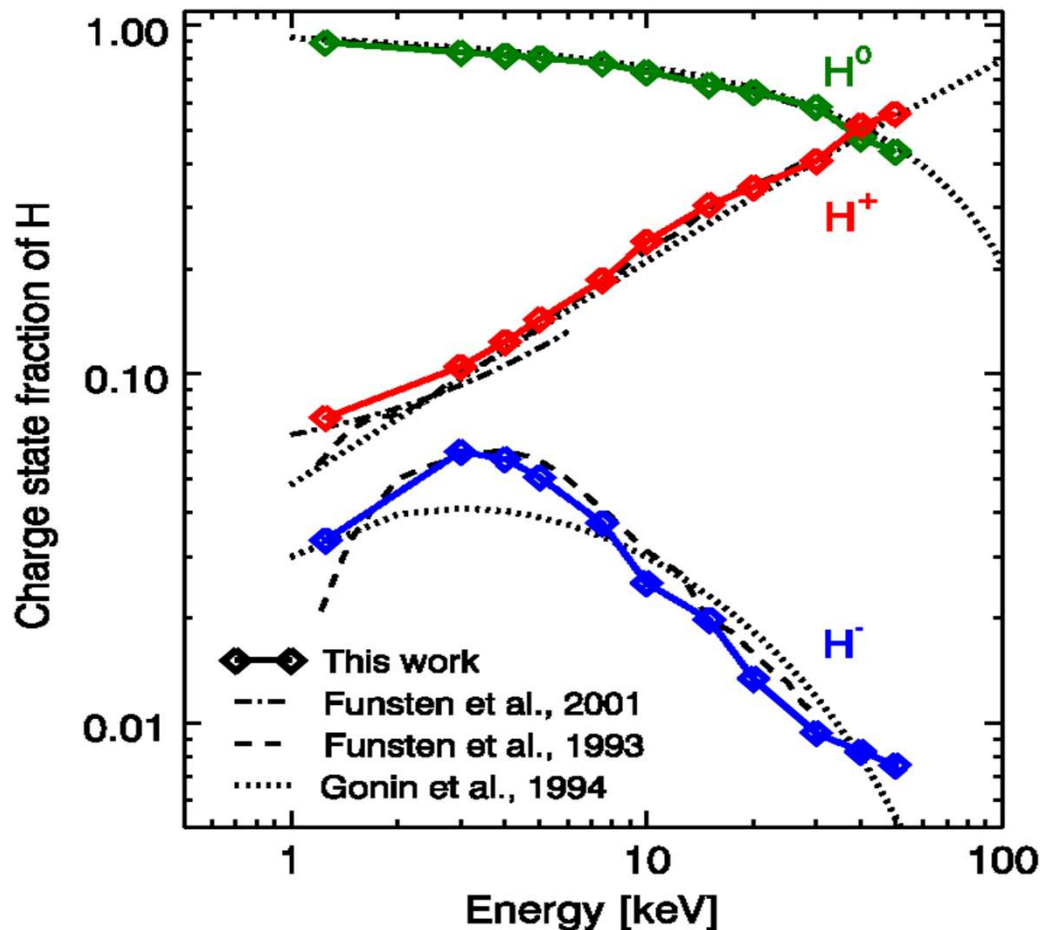


- Measure scattering angle half-width and use following relation to determine foil constant:
- Can assume the following* to relate foil constant to foil thickness:

$$E = \frac{k_F}{\psi_{1/2}}$$

$$t(\mu\text{g}/\text{cm}^2) = 0.08k_F(\text{keV deg})$$

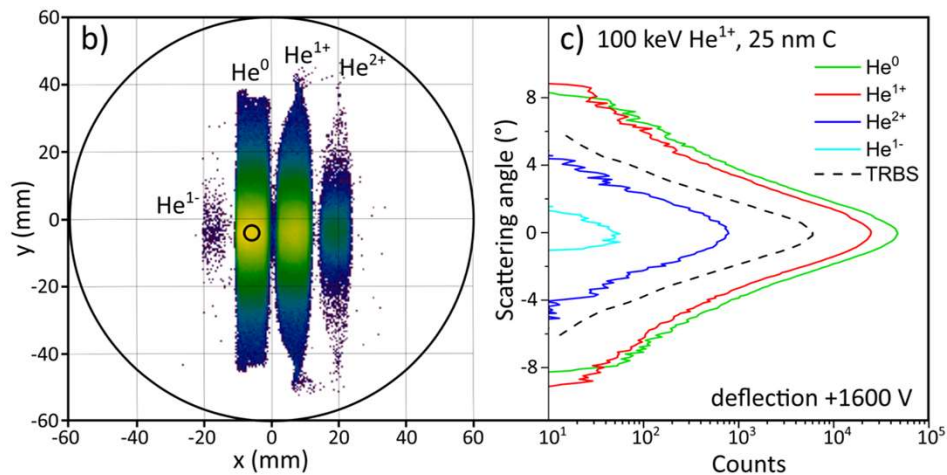
Previous Findings: Charge Fraction vs. Energy



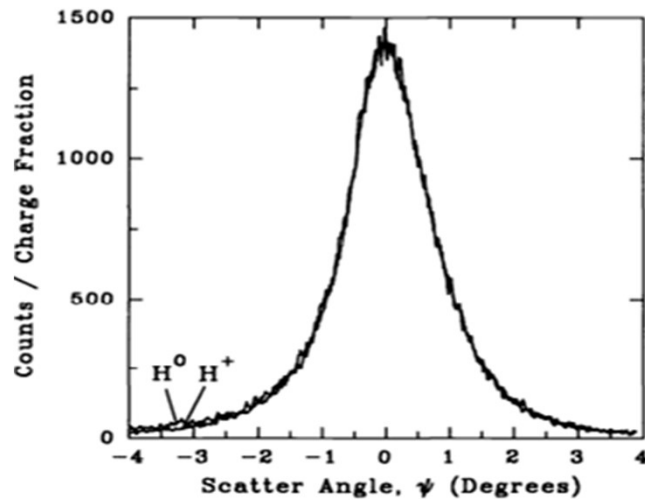
- Fraction of neutral atoms and negative ions decreases with energy
- Understanding that counts differ for different charge states is essential for analysis of their distributions
- Holenak et al. 2021 provides the same measurements for N, O and Ne isotopes

Allegrini et al. (2014)

Previous Findings: Counts vs Scattering Angle

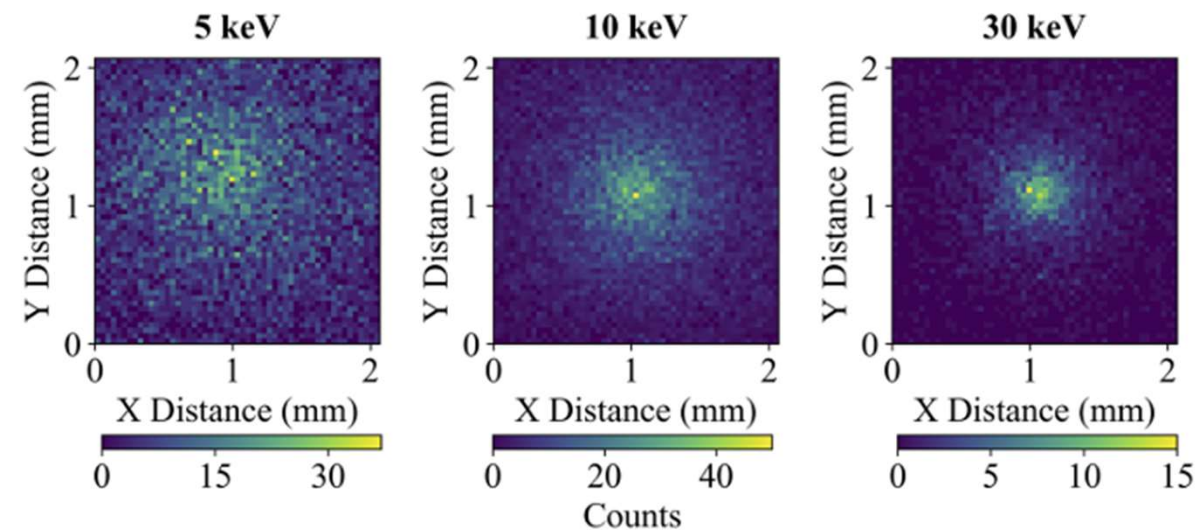


- Scattering can be modelled with a Kappa distribution
- Slight variations in this distribution reveal thin defects in the foil



Holenak et. al 2021; Funsten et al. 1993

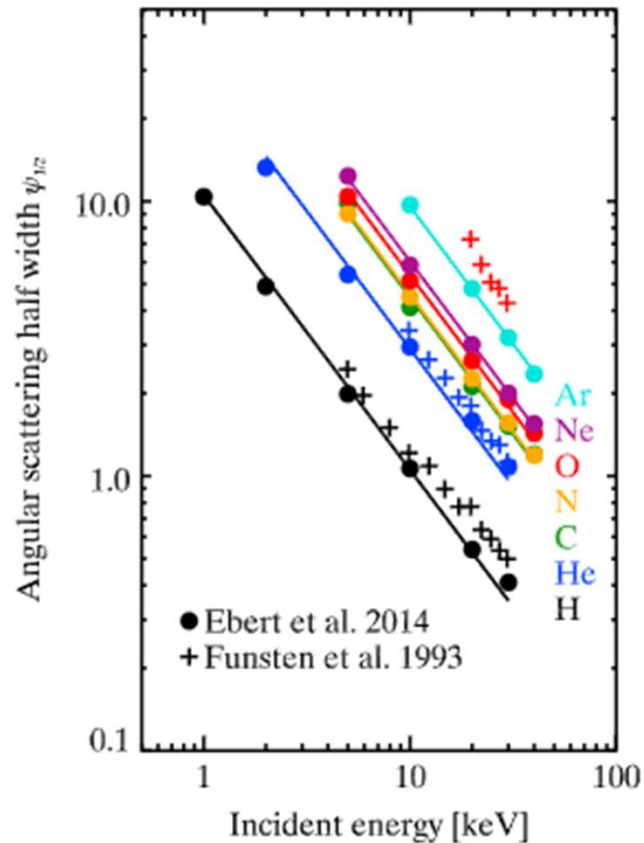
Previous Findings: Scattering Half-Width vs. Energy



- Scattering increases with lower energy ions
- FWHM will change depending on what energy ions we use

Vira et. al (2020)

Previous Findings: Scattering Half-Width vs. Energy

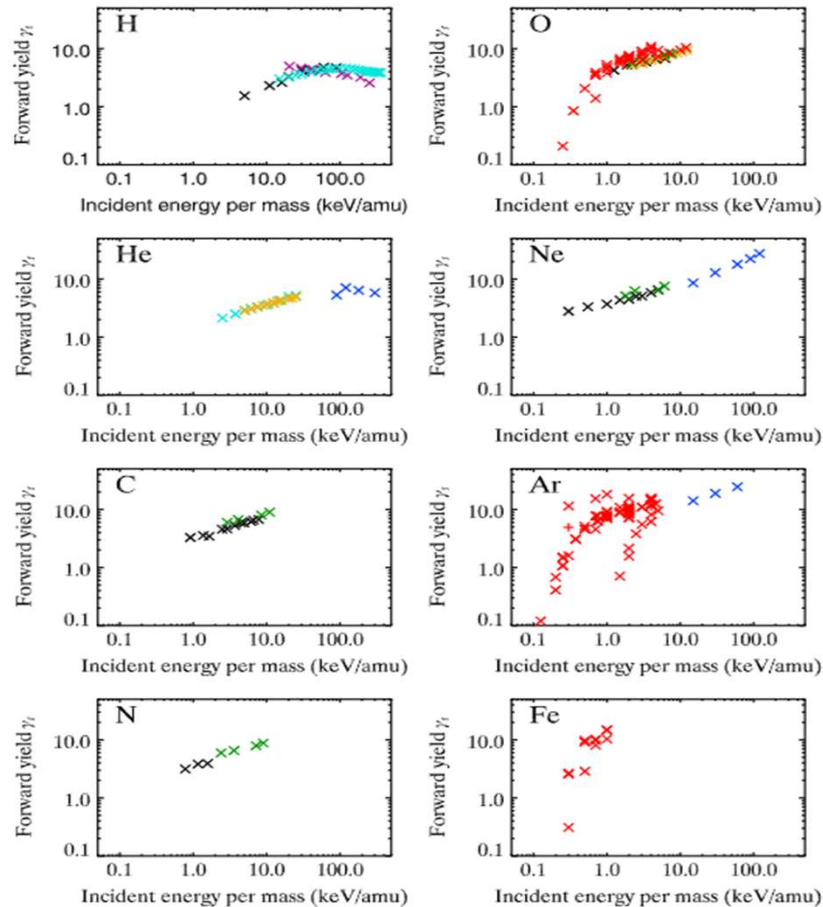


- Linear relationship between scattering angle half-width and energy of incident particle
- Heavier particles inherently have higher scattering angles

Allegrini et. al (2016)



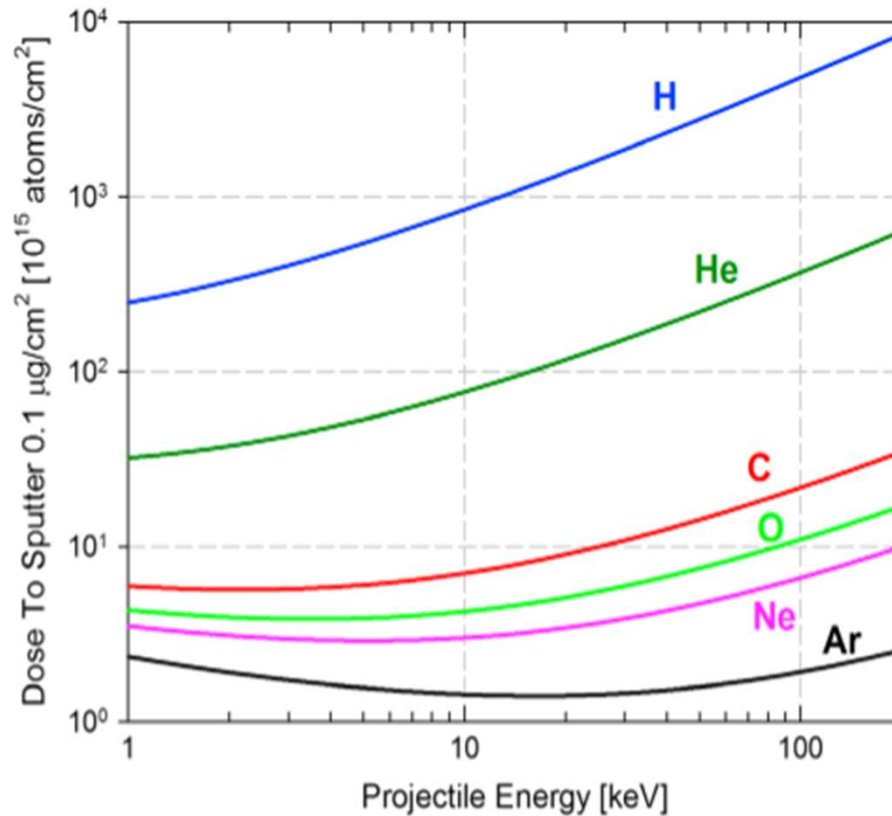
Previous Findings: Secondary Electron Emission



- Forward and backward secondary electron as a function of keV/amu has been determined
- Secondary electron energy distribution reveals peak energy of 2.1 eV, FWHM of 5.4 eV
- Projectile speed is ordering factor for secondary electron yield

* Allegrini et. al (2016)

Previous Findings: Foil Erosion Due to Sputtering



- Sputtering is an effect to worry about only on large scales and extended time periods. It would take solar wind at 1 keV/amu 7800 years to remove 0.1 μg/cm²*

*Allegri et. al (2016)



Preliminary Design Review

Science Goals and Conceptual Design



Basic Design Approach

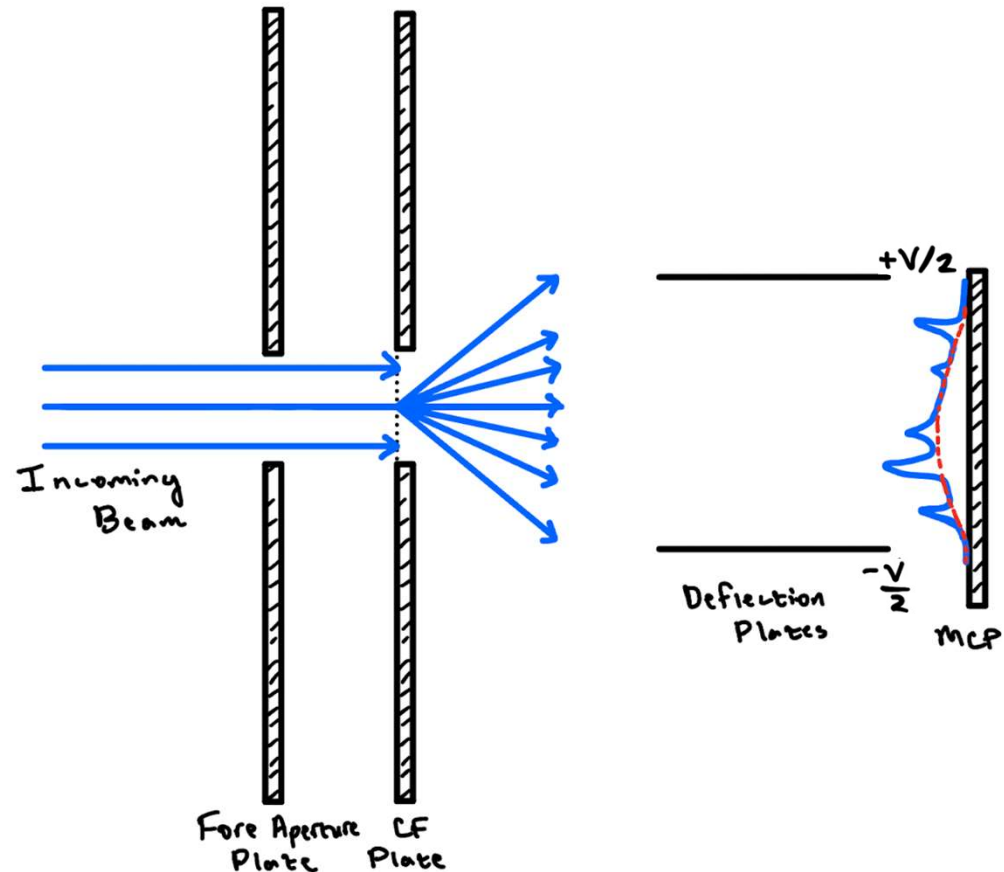
What is the simplest possible apparatus that will yield a science result?

What is the easiest thing we can add to that apparatus to produce another science result?

Design Stage 1

(1) Pinholes and extremely thin defects:

- Past studies* have produced surface maps of features but never differentiated between pinholes and thin defects
- Perhaps thin defects are significantly more effective than pinholes?

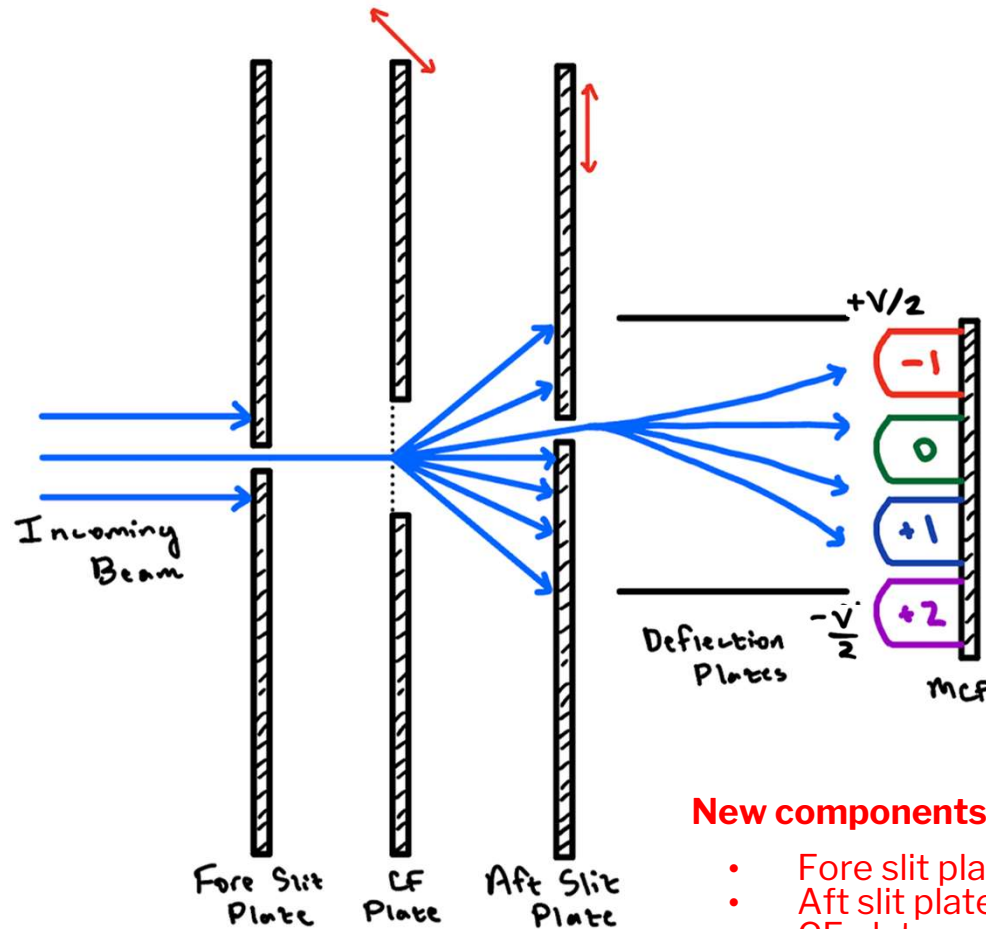


*e.g. Funsten et al 1992

Design Stage 2

Exit charge state partitioning versus (2) scattering angle and (3) scattering angle half-width:

- Past studies* have examined exit charge state partitioning versus energy, species, and exit velocity
- Perhaps the neutral fraction is greatest at the center of the scattering distribution, for example?



New components:

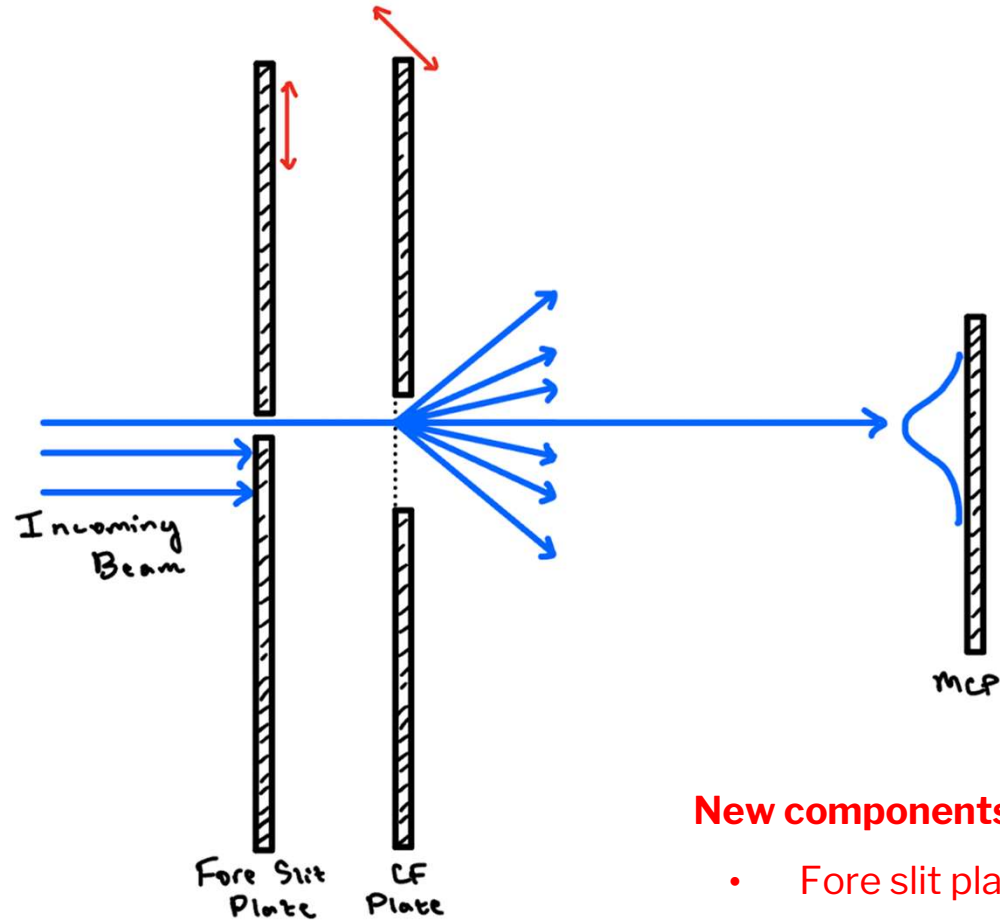
- Fore slit plate
- Aft slit plate
- CF plate and aft slit plate actuators

*e.g., Allegrini et al. 2014; Vira et al. 2020; Holenak et al. 2021

Design Stage 3a

(4) Variation of thickness across the surface of the foil:

- Past studies* have determined bulk thickness for various foils but never locally across a surface

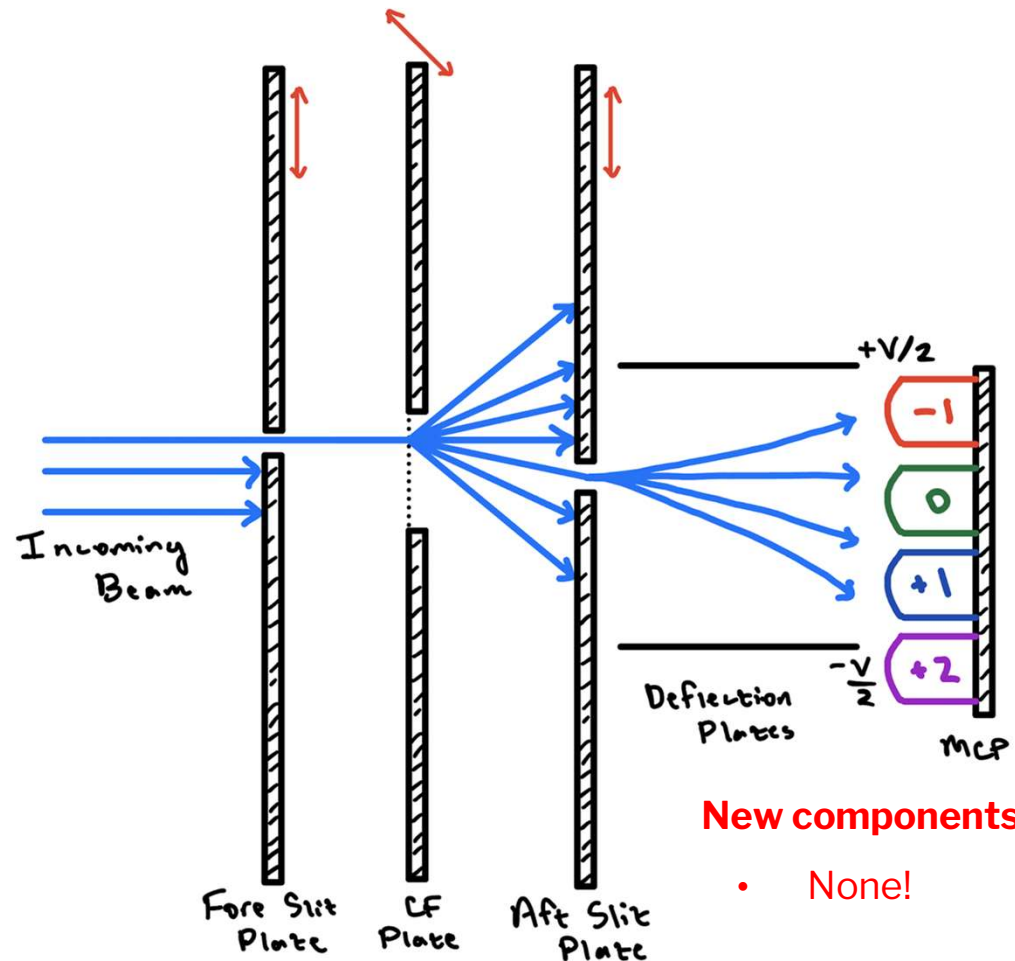


*e.g., Allegrini et al. 2016; Fhert et al. 2014; Meyer 1970

Design Stage 3b

(5) How does local thickness influence exit charge state partitioning for a given scattering angle?

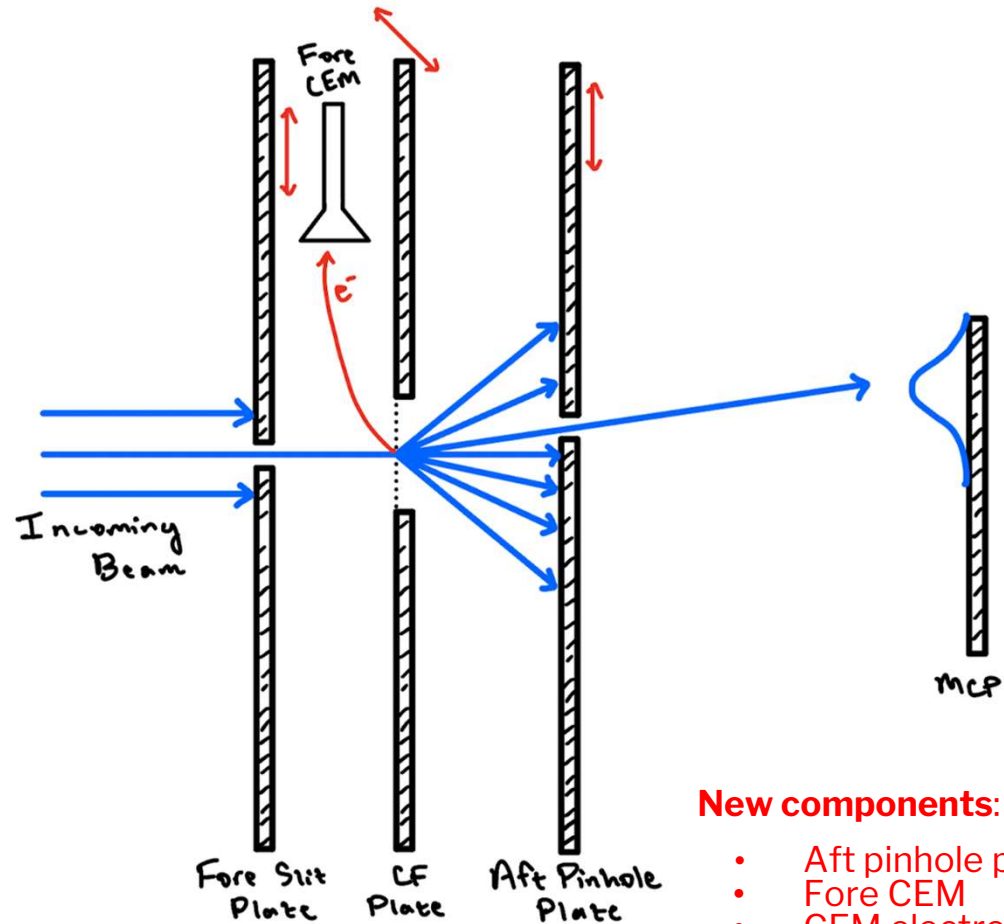
- Combining the answers to questions (2)–(4) gives this question, which has yet to be asked by other scientists
- Indirectly tells us how exit charge state partitioning varies with thickness and across the surface of a given carbon foil



Design Stage 4

(6) How is the rate of secondary electron emissions related to angular scattering angle?

- Past studies* have examined secondary electron yield versus energy and fluence but never scattering angle
- Perhaps ions with large scattering angles produce more secondary electrons since they pass through more carbon?



New components:

- Aft pinhole plate
- Fore CEM
- CEM electronics
- Coincidence electronics

*e.g. Ritzau et al 1998

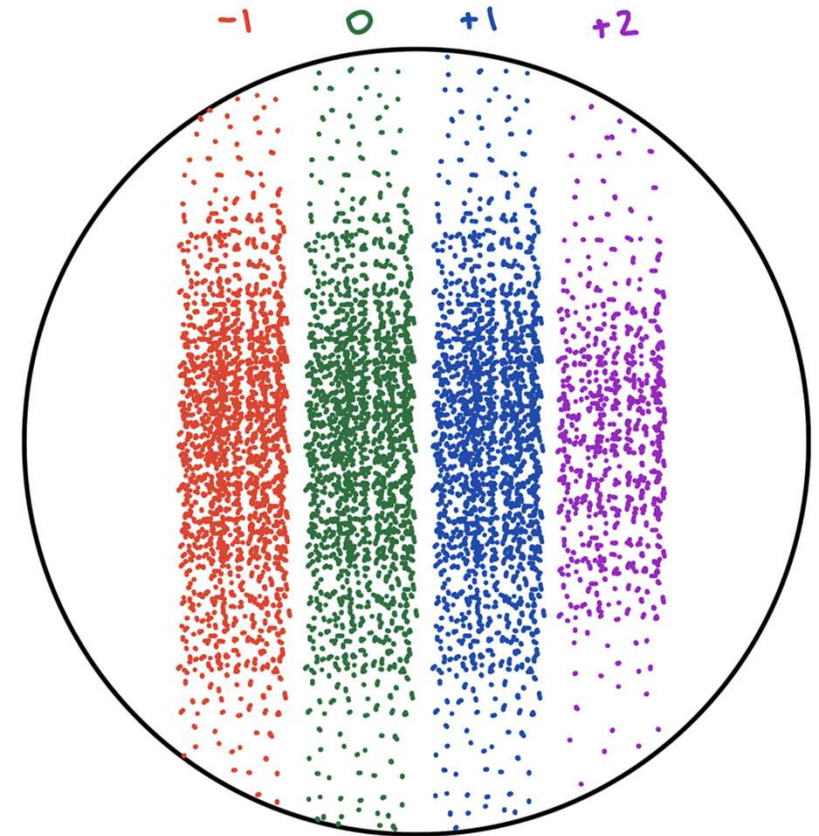


Preliminary Design Review

Apparatus Design

Design Specifications: Basic Requirements

- Most incoming ions should strike MCP
- Charge state populations should be separated on MCP
- High count rates $\rightarrow$ good statistics
- Minimal E-field distortion/fringe effects



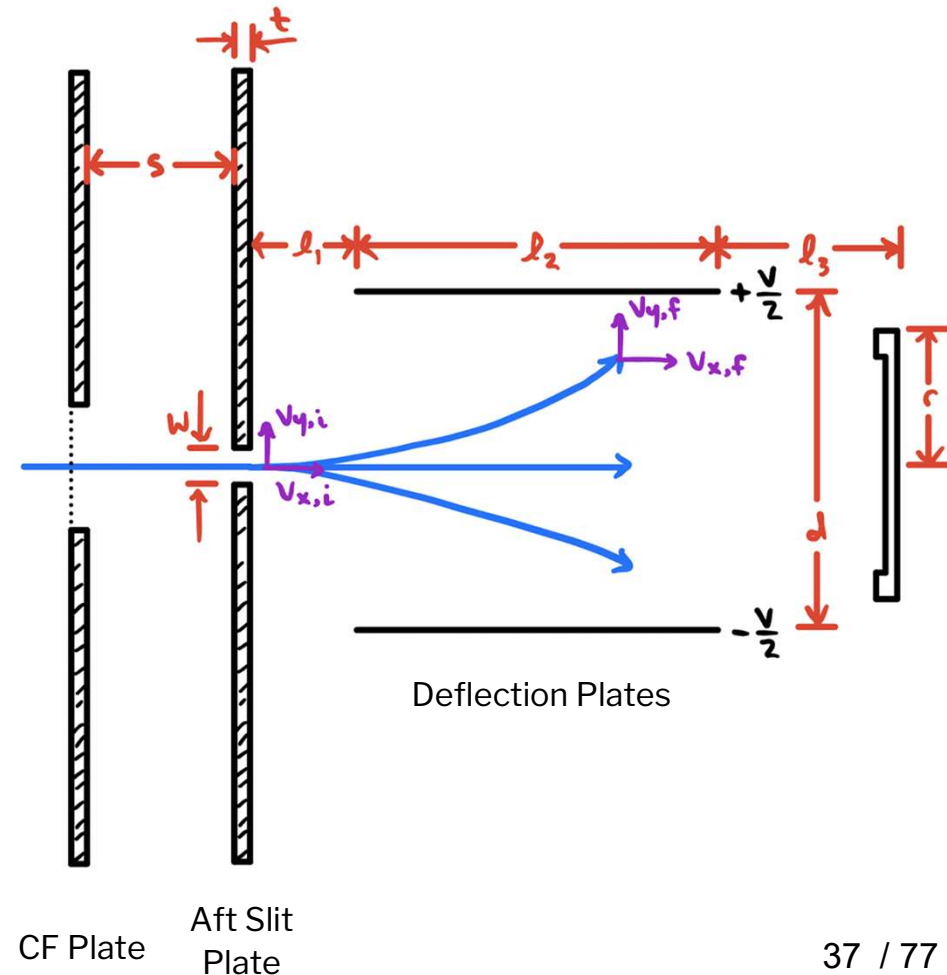
Design Specifications: Deflection Constraints

Consider Δy of an ion after passing through the following three segments:

1. Aft slit plate to deflection plates (Δy_1)
2. Deflection plates (Δy_2)
3. Deflection plates to MCP (Δy_3)

Then, the total deflection is given by

$$\Delta y = \Delta y_1 + \Delta y_2 + \Delta y_3$$



Design Specifications: Deflection Constraints

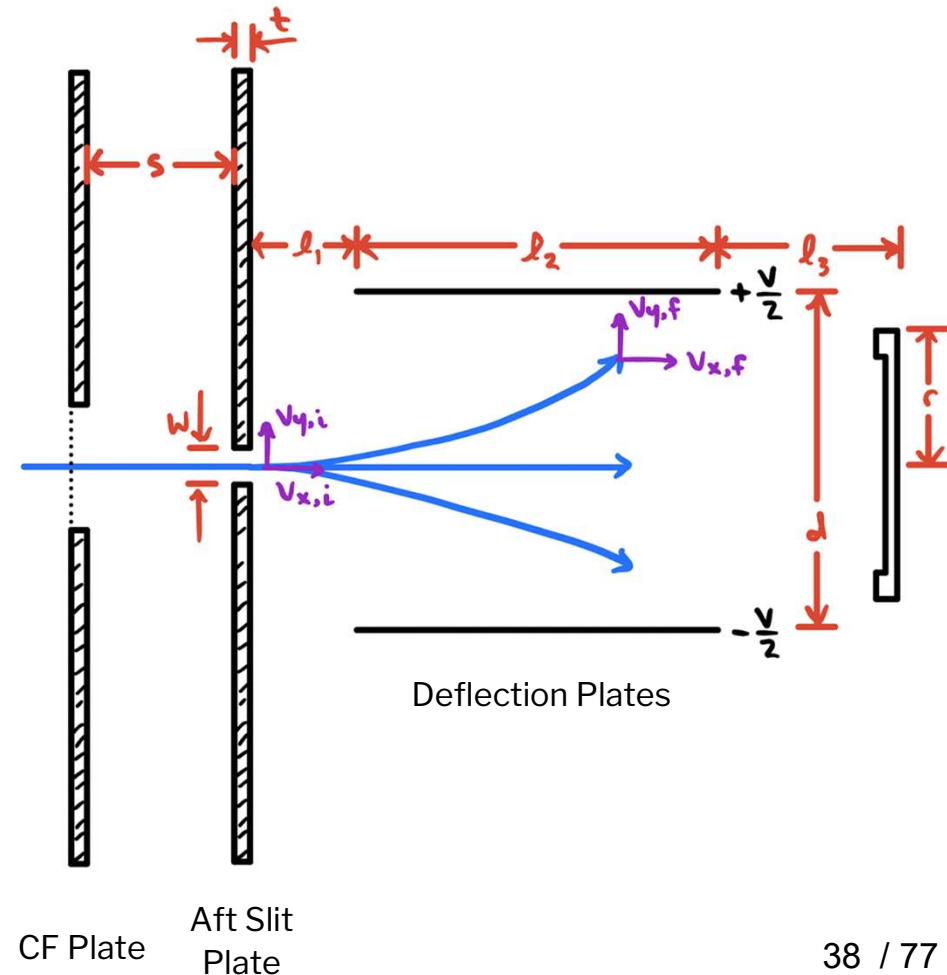
Consider Δy of an ion after passing through the following three segments:

1. Aft slit plate to deflection plates (Δy_1)
2. Deflection plates (Δy_2)
3. Deflection plates to MCP (Δy_3)

Then, the total deflection is given by

$$\Delta y = \Delta y_1 + \Delta y_2 + \Delta y_3$$

$$\Delta y = \frac{w}{t}(\ell_1 + \ell_2 + \ell_3) + \frac{Vq\ell_2}{2Ed} \left(1 + \frac{w^2}{t^2} \right) \left(\frac{\ell_2}{2} + \ell_3 \right)$$





Design Specifications: Deflection Constraints

With this expression, we can:

1. Take $\Delta y < r$ to ensure all
deflected ions strike the MCP

$$r > \alpha(\ell_1 + \ell_2 + \ell_3) + \frac{Vq\ell_2}{2Ed} (1 + \alpha^2) \left(\frac{\ell_2}{2} + \ell_3 \right)$$

2. Take $\Delta y < d/2$ and $\ell_3 = 0$ to ensure
no deflected ions strike the plates

$$\frac{d}{2} > \alpha(\ell_1 + \ell_2) + \frac{Vq\ell_2^2}{4Ed} (1 + \alpha^2)$$

3. Take $\Delta y_{q=+1} > \Delta y_{q=0}$ to ensure that
charge state populations are
separated

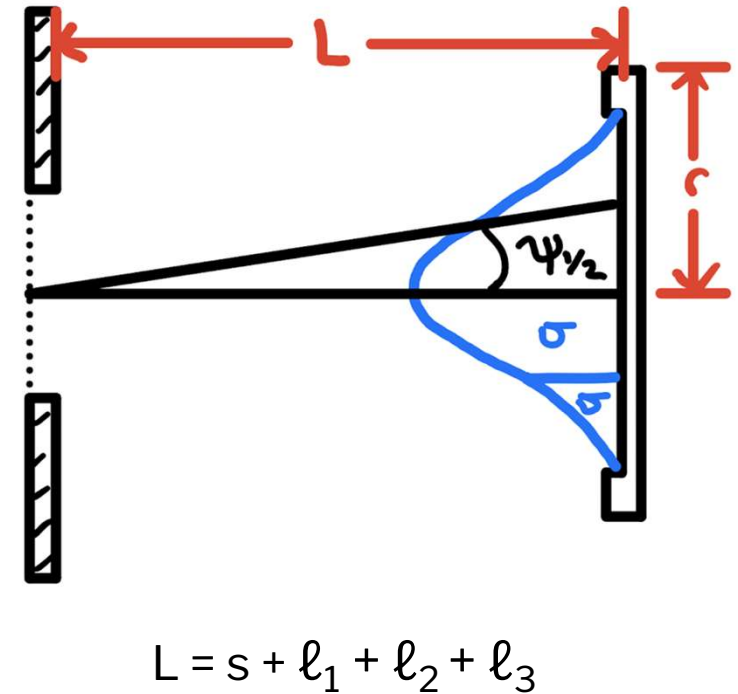
$$\frac{V|q|\ell_2}{2Ed} (1 + \alpha^2) \left(\frac{\ell_2}{2} + \ell_3 \right) > 2\alpha(\ell_1 + \ell_2 + \ell_3)$$

Design Specifications: Scattering Constraint

To ensure that the majority (95%) of the scattered ions hit the MCP, we can require that $2\sigma < r$:

$$\psi_{1/2} < \tan^{-1} \left(\sqrt{\frac{\ln 2}{2}} \frac{r}{s + \ell_1 + \ell_2 + \ell_3} \right)$$

where $\psi_{1/2}$ is the scattering angle half-width.





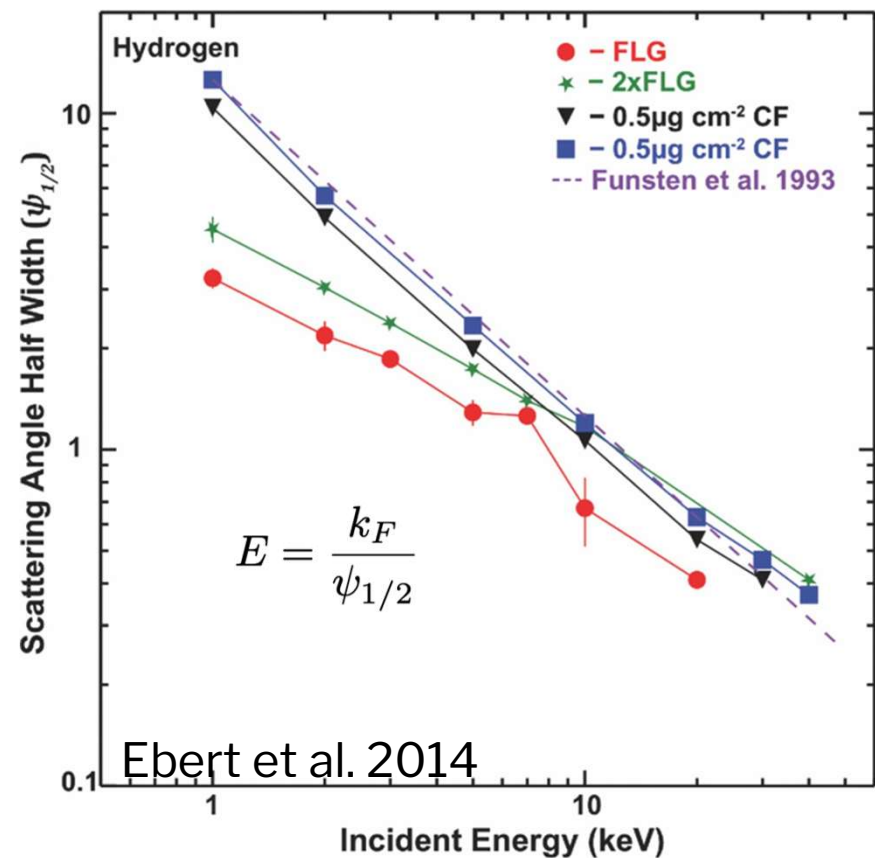
Design Specifications: Scattering Constraint

$$\psi_{1/2} < \tan^{-1} \left(\sqrt{\frac{\ln 2}{2}} \frac{r}{s + \ell_1 + \ell_2 + \ell_3} \right)$$

$$E = \frac{k_F}{\psi_{1/2}}$$

$$t(\mu\text{g}/\text{cm}^2) = 0.08 k_F (\text{keV deg})^*$$

$$E > \frac{12.5t}{\tan^{-1} \left(\sqrt{\frac{\ln 2}{2}} \frac{r}{(s + \ell_1 + \ell_2 + \ell_3)} \right)}$$



* $\propto$ Funsten et al 1993



Design Specifications: Constraint Overview

1. Deflected ions strike MCP

$$r > \alpha(\ell_1 + \ell_2 + \ell_3) + \frac{Vq\ell_2}{2Ed} (1 + \alpha^2) \left(\frac{\ell_2}{2} + \ell_3 \right)$$

2. Deflected ions miss DPs

$$\frac{d}{2} > \alpha(\ell_1 + \ell_2) + \frac{Vq\ell_2^2}{4Ed} (1 + \alpha^2)$$

3. Charge state pops. separated

$$\frac{V|q|\ell_2}{2Ed} (1 + \alpha^2) \left(\frac{\ell_2}{2} + \ell_3 \right) > 2\alpha(\ell_1 + \ell_2 + \ell_3)$$

4. Majority of angular scattering distribution strikes MCP

$$E > \frac{12.5t}{\tan^{-1} \left(\sqrt{\frac{\ln 2}{2}} \frac{r}{(s + \ell_1 + \ell_2 + \ell_3)} \right)}$$



Design Specifications: Possible Parameters

Parameter	Value
E	650 – 1250 eV
V	700 – 2500 V
q	$(-1e) - (+2e)$
r	20 mm
s	20 mm
d	12 mm
w	1 mm
t	4 mm
ℓ_1	2 mm
ℓ_2	8 mm
ℓ_3	12 mm

Design Specifications: SIMION

$E = 650 \text{ eV}$

$V = 700 \text{ V}$

$\psi_{1/2} = 15.4^\circ$

Protons

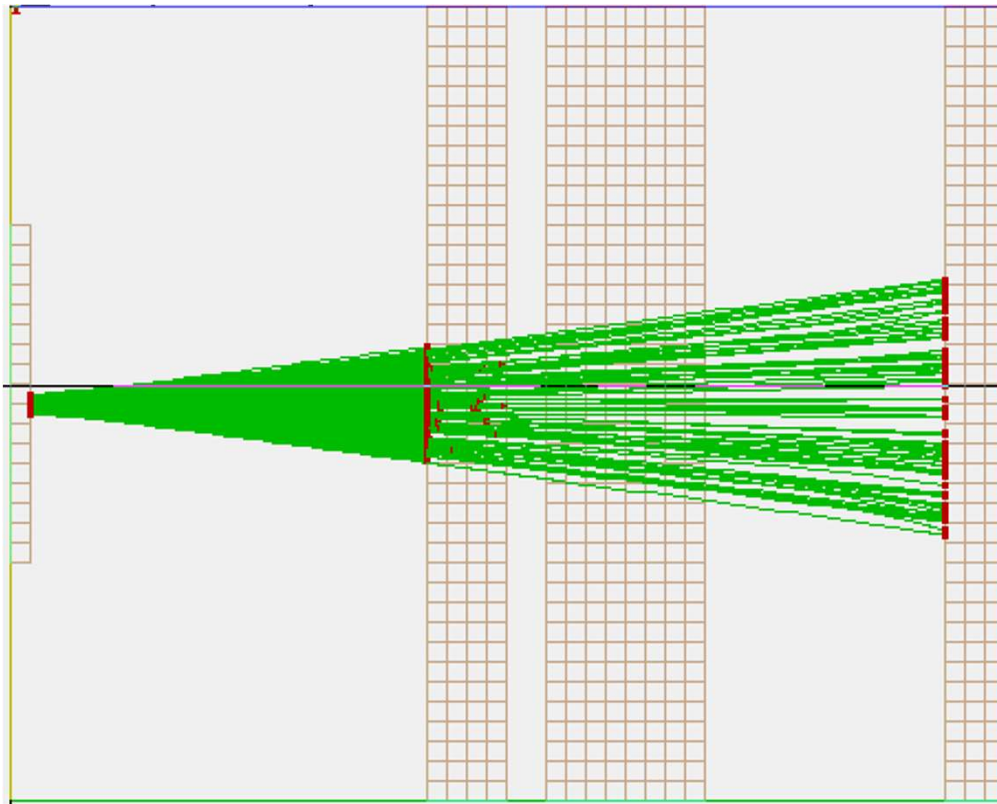
$E = 1250 \text{ eV}$

$V = 1400 \text{ V}$

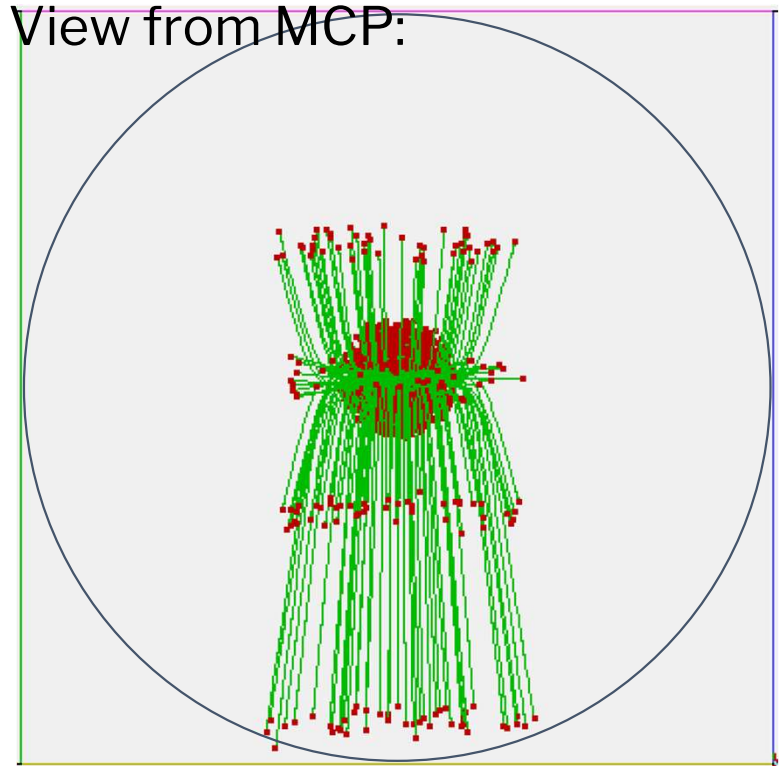
$\psi_{1/2} = 8^\circ$

Protons

Design Specifications: SIMION



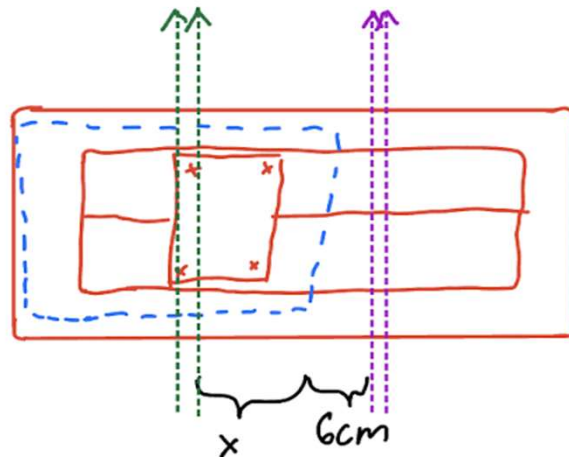
View from MCP:



Translational Stage Considerations

Limiting dimensions:

- Device moved into beam line
- Device moved out of beam line
- Device within weight and torque tolerances of stages



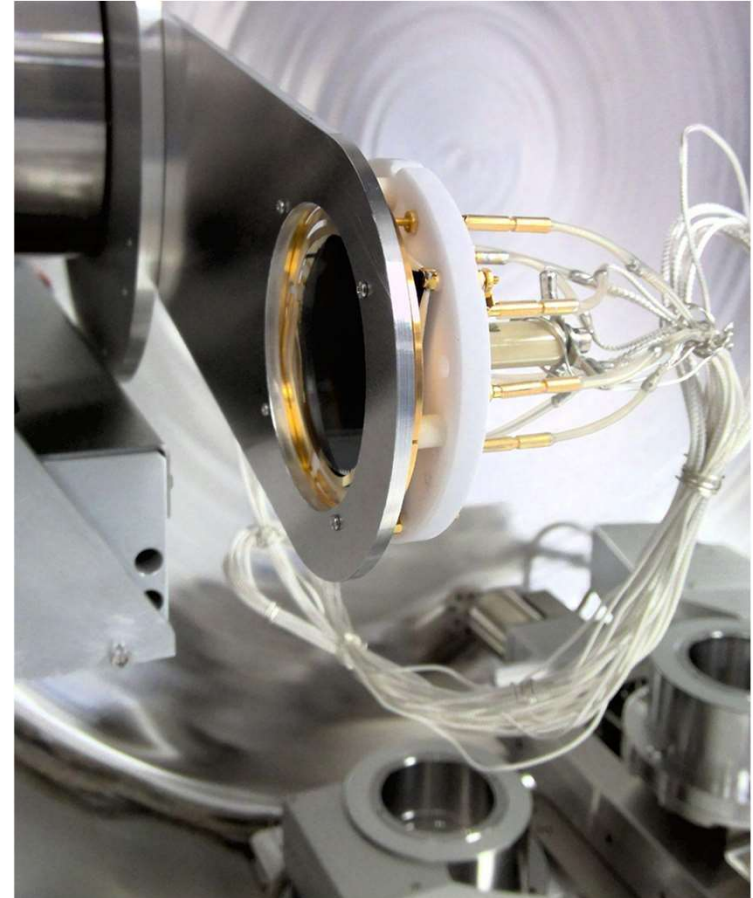


Preliminary Design Review

Main Apparatus Build Process

Parts from Lab: IMCP

- 1. Quantar 3394A
- 1. Diameter : 73.7mm
- 1. Thickness: 1.985mm
- 1. Active diameter: 40mm





Parts from Lab: Carbon Foil

1. Carbon foil diameter: 18mm
2. Carbon foil frame diameter: 38mm
3. Plate dimensions: 50 x 50mm
4. Thickness: 1mm
5. Weight: 5.72g

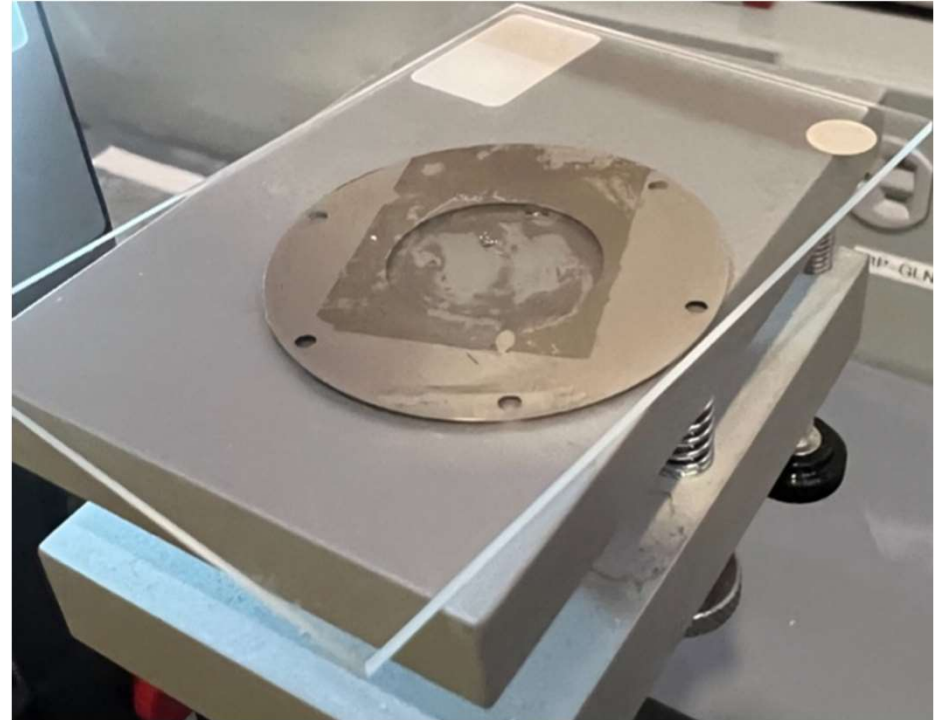
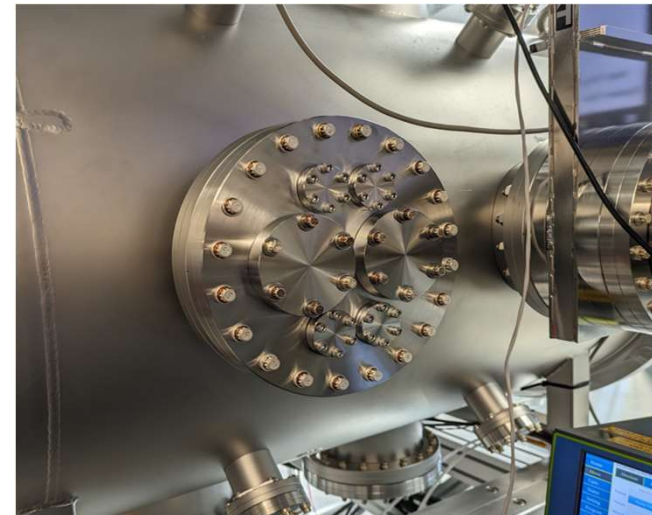
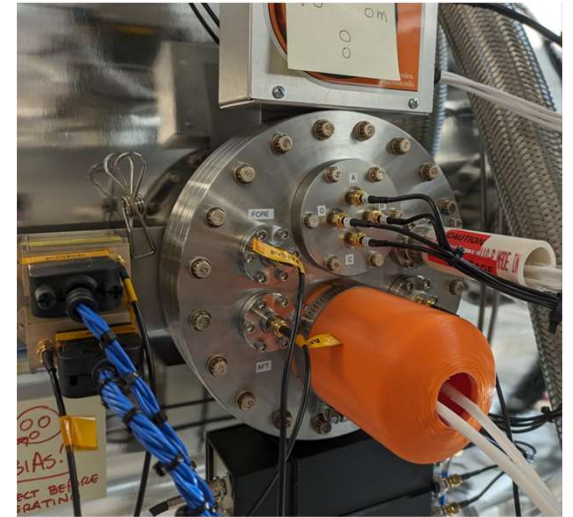


Photo credits: Joey Avallone

Parts from Lab/Order: Wiring

1. HV feedthroughs (SHV-5 or SHV-10):
 - a. Voltage supply for MCP
 - b. Voltage supply for the deflection plates
 - c. Actuators
2. Signal feedthroughs (DB-25 or DB-9)
 - a. Data connection from MCP
3. 3D print protective can
4. Connecting to chamber: CF Flanges
5. Wires and electronics: TRD



Parts To Be Ordered: Actuators I



N-331 PICMAWalk Walking Drive

- Vacuum version 10^{-9} hPa
- Travel distance 25-100mm
- 10mm/s travel time
- 20nm resolution
- Quote Pending



Faulhaber Piezo Linear Motor

- Vacuum compatible 10^{-7} torr (materials list pending)
- Travel distance 80mm
- 10mm/s travel time
- \$1376

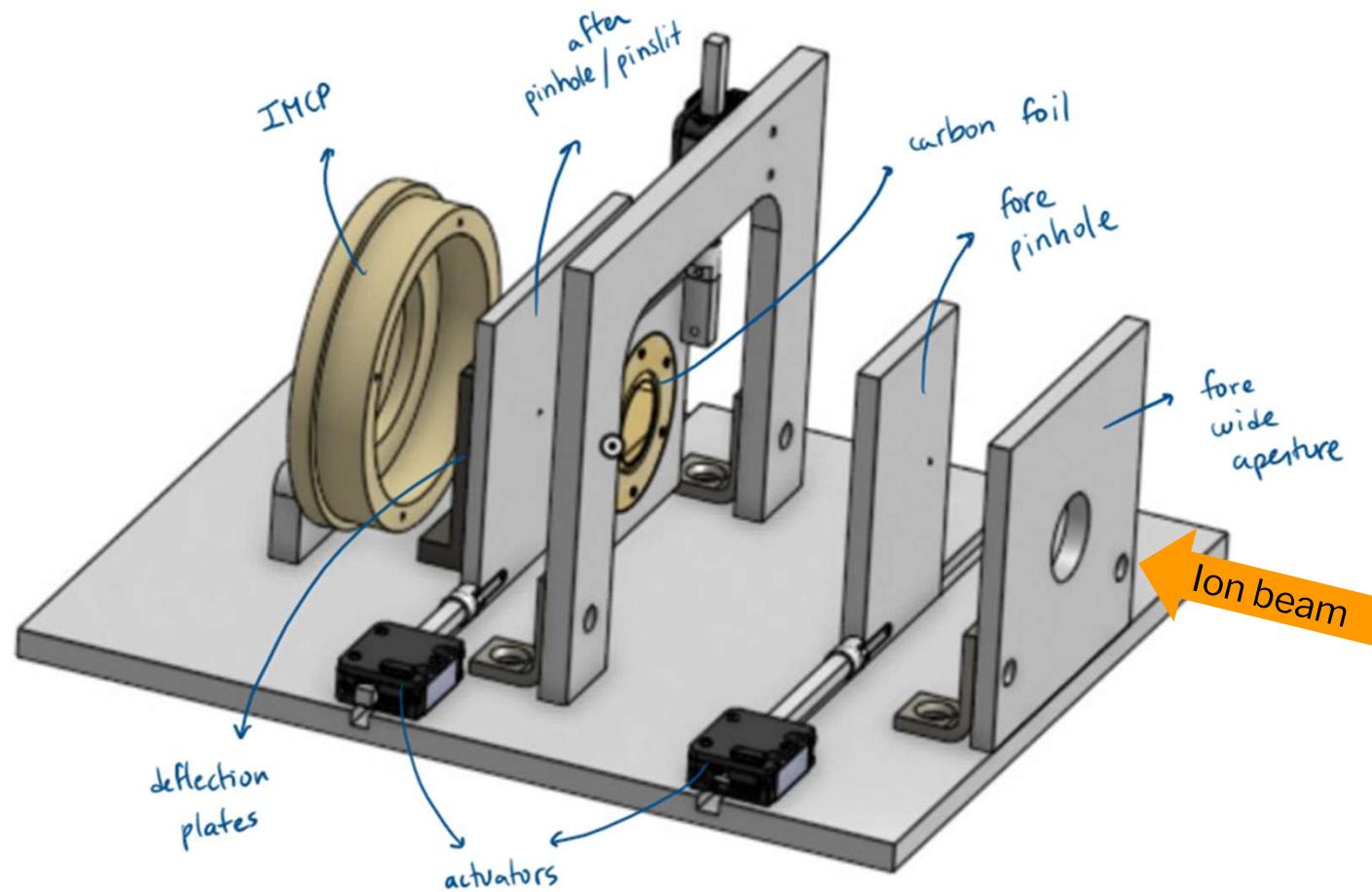
Parts To Be Ordered: Actuators II



Piezoelectric Inertia Actuator (Thorlabs)

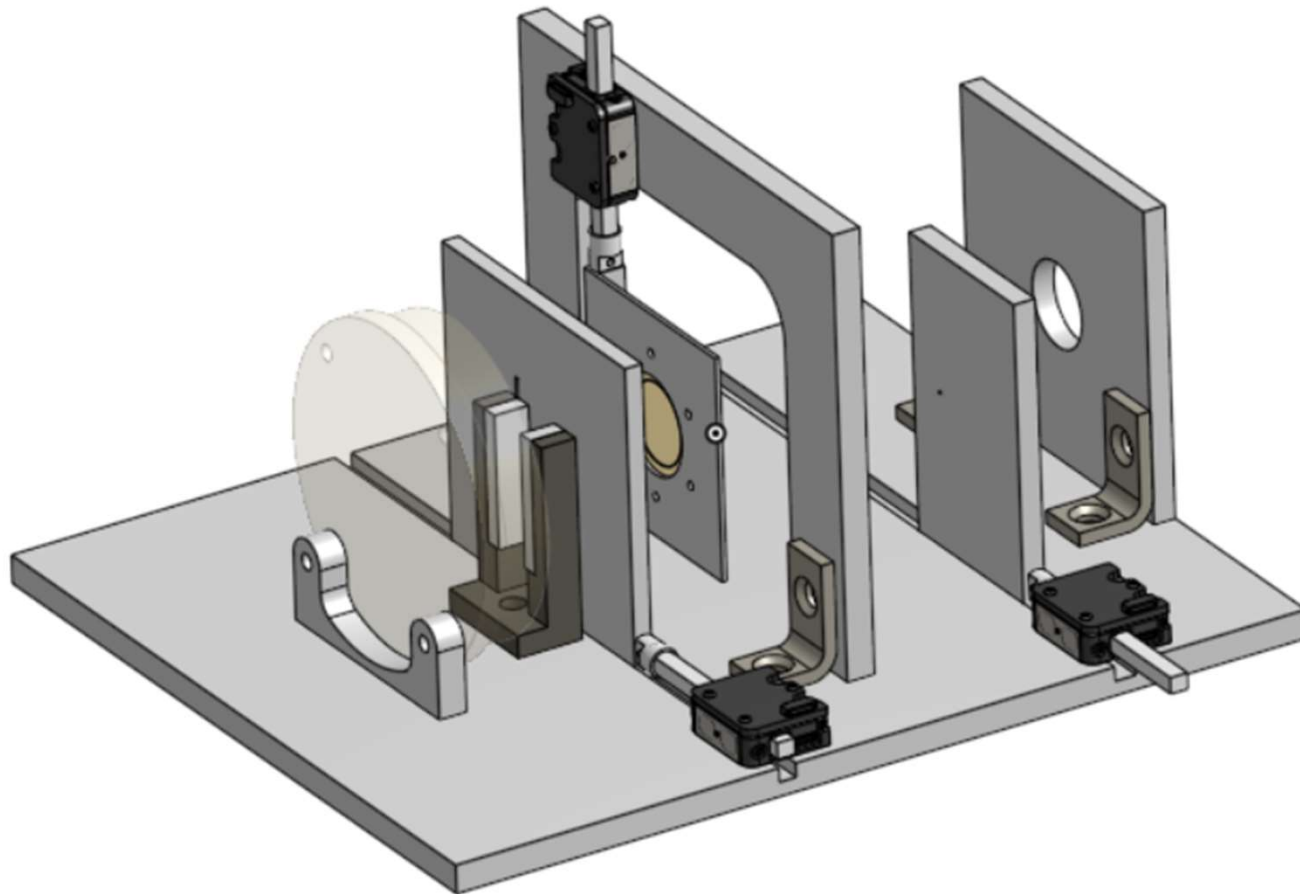
- Vacuum Compatible 10^{-6} torr
- 13mm travel range
- 20 nm resolution
- 2mm/minute travel time
- \$873
- Round drive shaft

Instrument Assembly



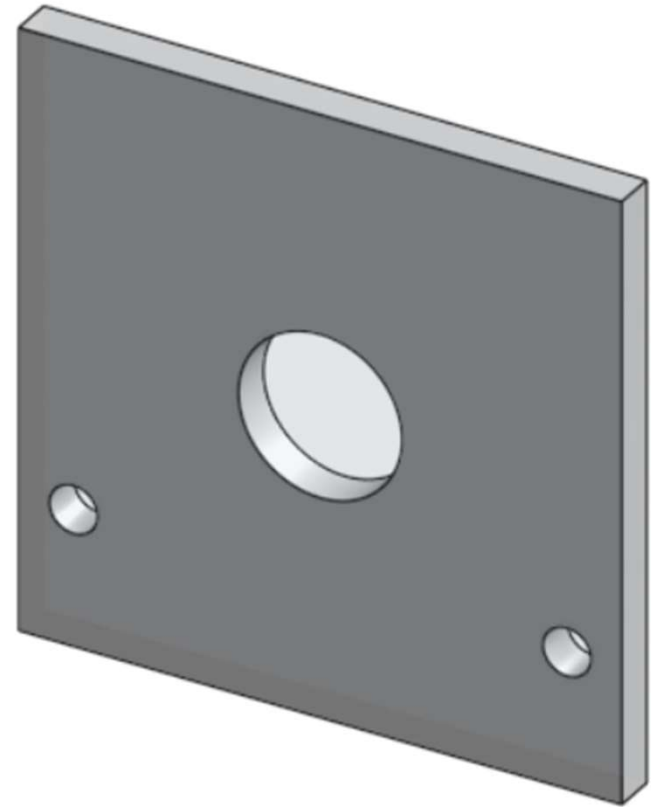
CAD design

Instrument Assembly



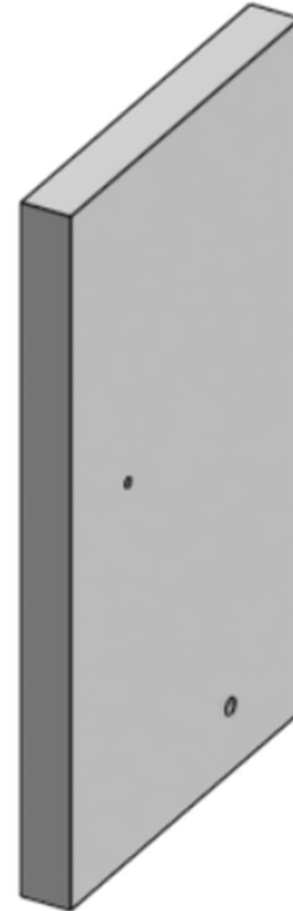
Machined Parts: Aperture plate

1. Plate dimensions: 70 x 70mm
2. Plate thickness: 5mm
3. Aperture diameter: 19mm
4. Material : Al6061



Machined Parts: Fore pinhole

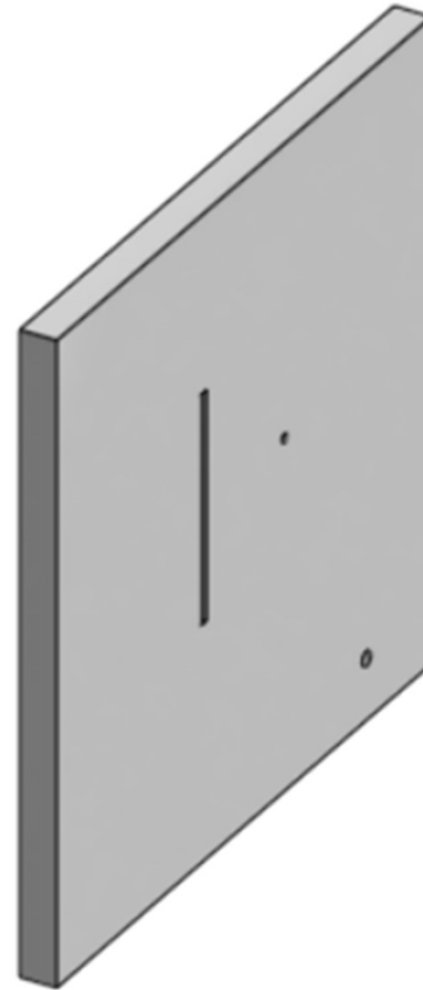
1. Plate dimensions: 40 x 70mm
2. Plate thickness: 5mm
3. Pinhole diameter: 1mm
4. Material: Al6061





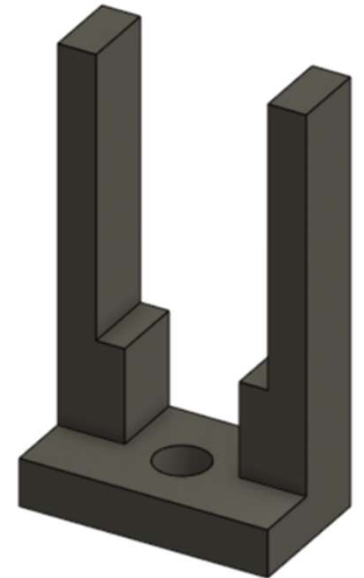
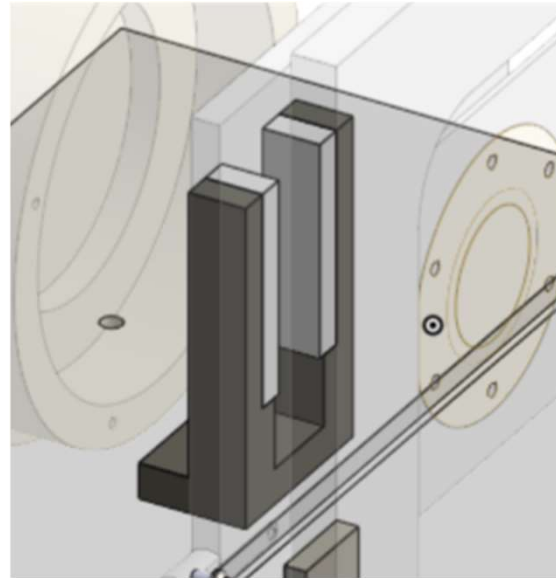
Machined Parts: Aft Pinhole/pinslit

1. Plate:
 - a. Dimensions: 70 x 70mm
 - b. Thickness: 4mm
2. Pinhole:
 - a. Diameter: 1mm
3. Pinslit (vertical):
 - a. Width: 1mm
 - b. Length: 25mm
4. Material: Al6061



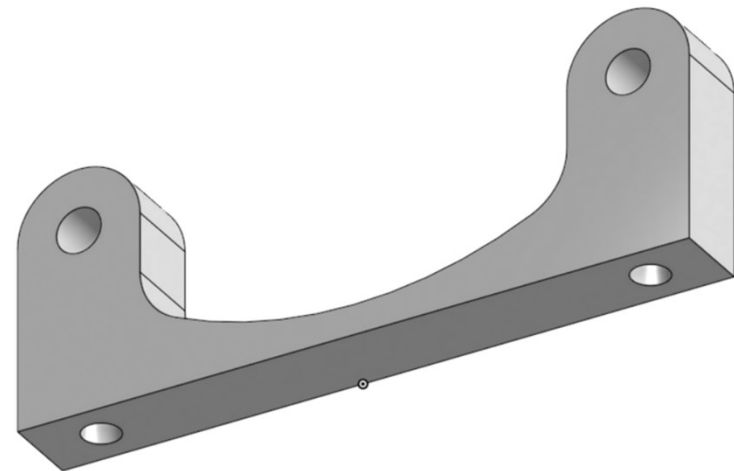
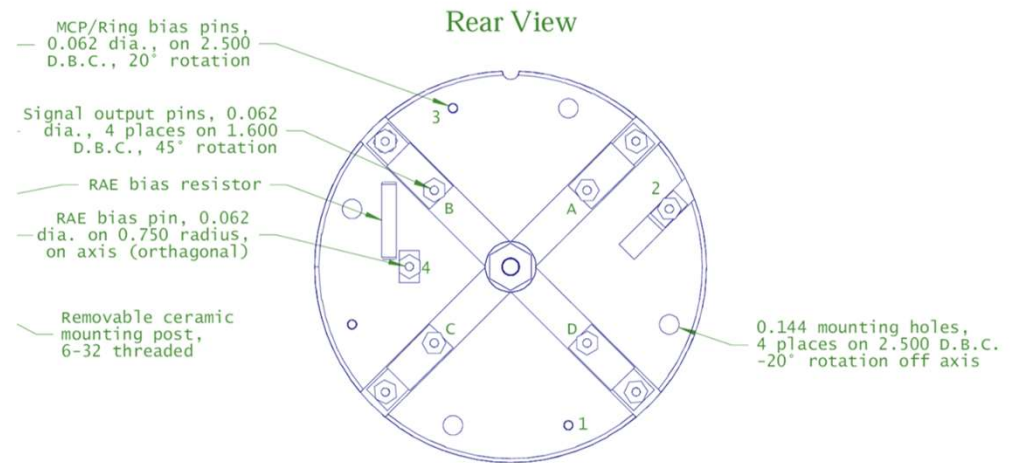
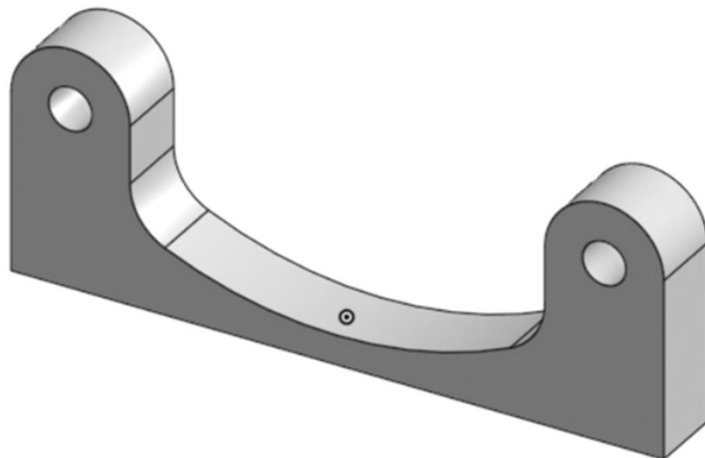
Machined Parts: Deflection Plates

1. Plate length: 8 mm
2. Plate thickness: 4 mm
3. Separation between plates: 12 mm
4. Plate material: Al6061 (tentatively)
5. Mount material: PTFE

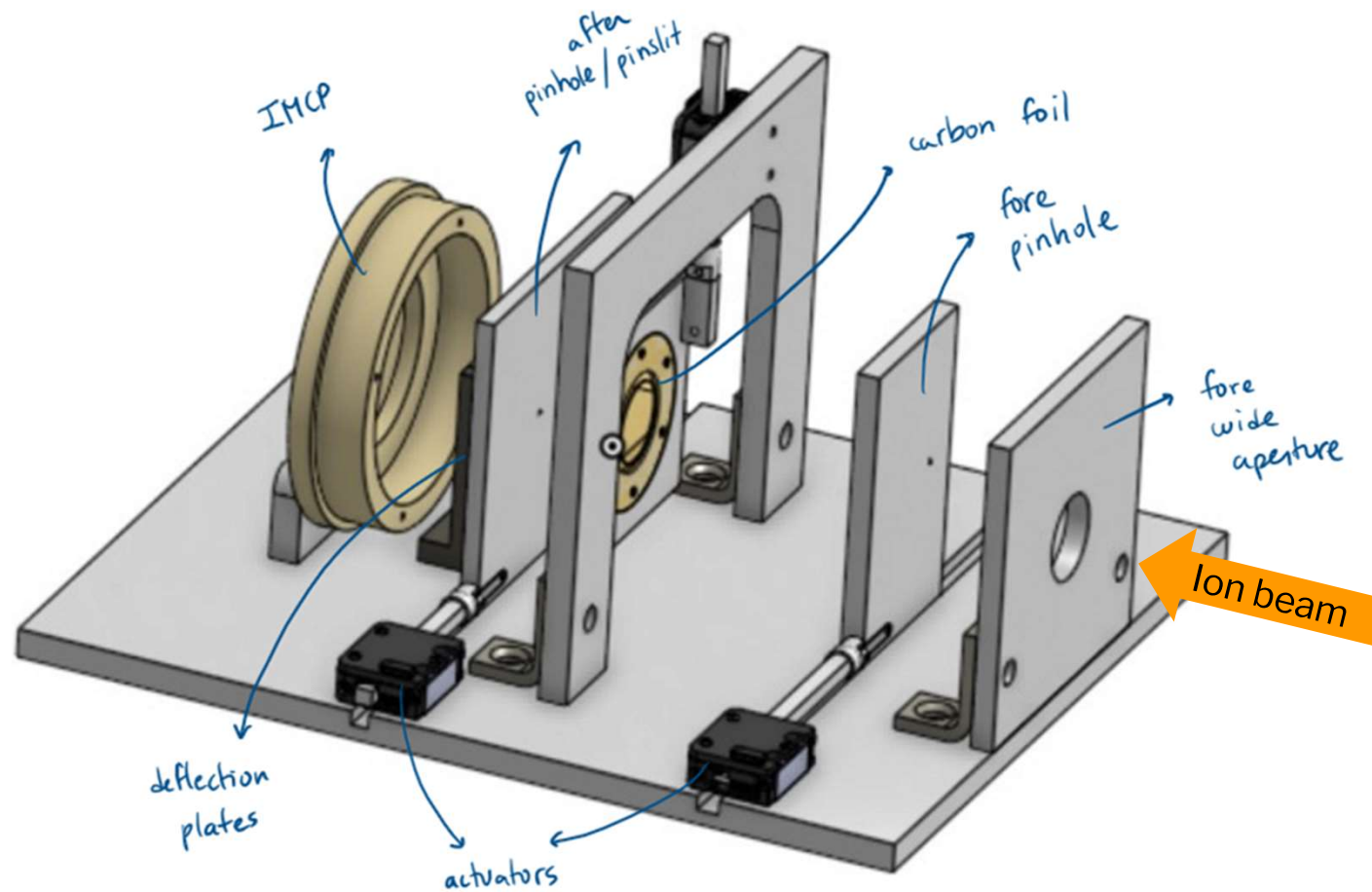


Machined Parts: IMCP Mount

1. Mount width: 54.9mm
2. Mount thickness: 6.35mm
3. Mount screw height: 15mm
4. Plate material: Al6061

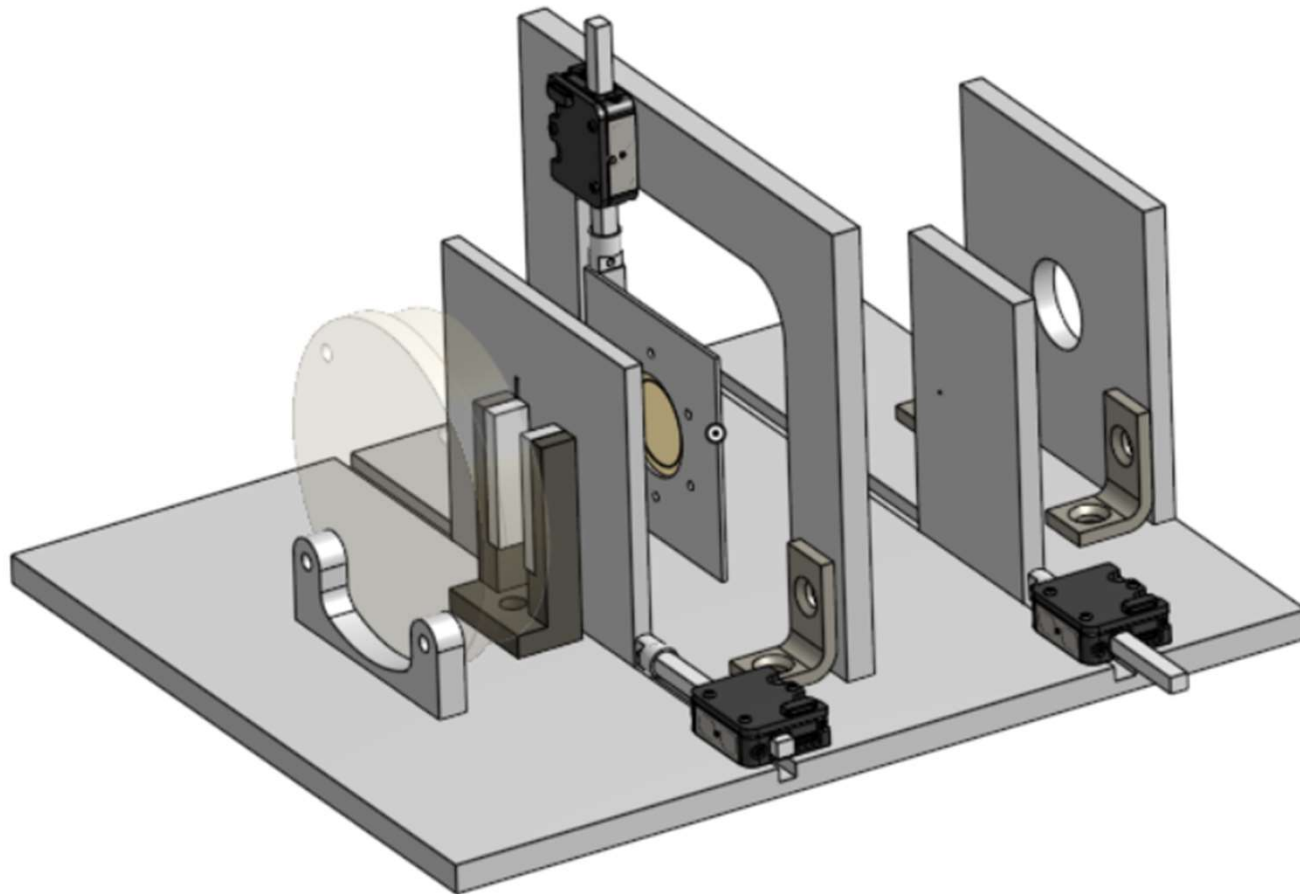


Instrument Assembly



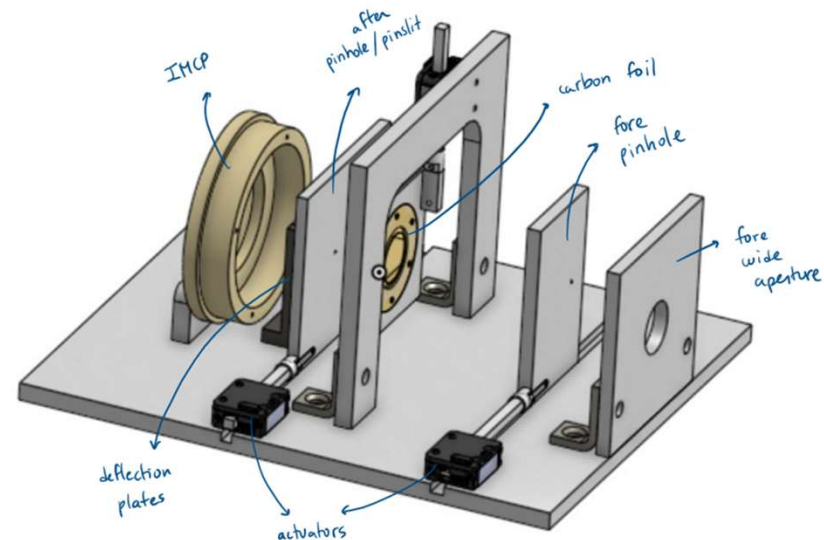
CAD design

Instrument Assembly



Instrument Assembly

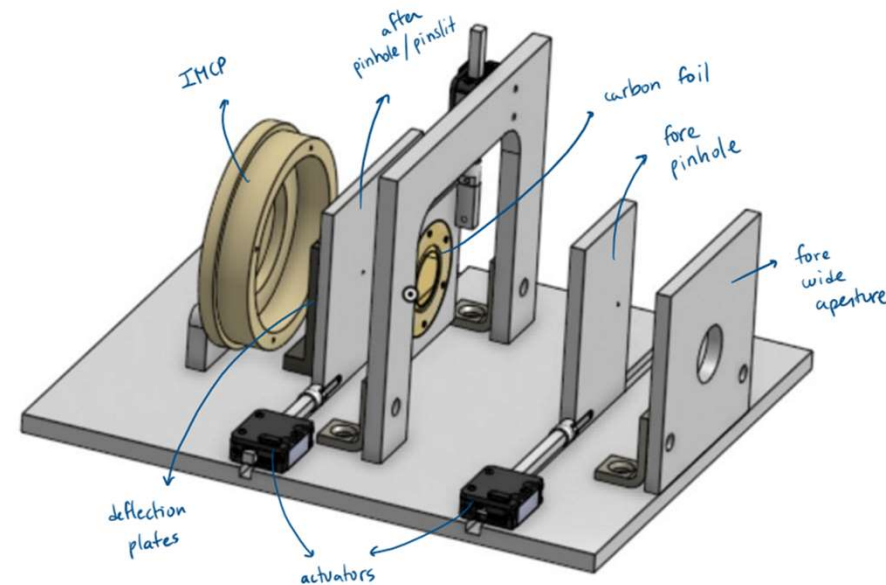
- Mounted on a rectangular plate for easier access to components. Will also be made of 6061 Al
- Dimensions: 200 mm x 170 mm
- Base frame has rails where the fore pinhole and after pinhole/slit are to help with alignment.
- Additional frame to support Carbon Foil actuator.



CAD design

Instrument Assembly

- Attachment to base plate with brackets, screws:
 - CF actuator mount, fore aperture
 - Stainless Steel Corner Bracket ([Link](#))
 - Stainless Steel Flat Head Blunt Screw ([Link](#))
 - Actuators - TBD
 - Voltage plate mount, IMCP mount - TBD
 - Determine threaded hole diameters



CAD design





Interfacing With the Device

1. MCP: interface will be similar to previous examples done in lab (documentation needed)
2. Deflector Plates: High Voltage(relative to device ground) line that can be adjusted from outside the chamber
3. Actuators: Implementation will depend on specific device chosen, but we're expecting likely 4-8 lines of data connections needed
4. Device Ground: The grounded voltage (likely the baseplate of the device) that will act as a ground that all other voltages are compared against.
5. Linear and Rotational Staging: This is currently being worked on in the lab, and should be a solved problem for us already



Preliminary Design Review

Challenges



Anticipated Challenges

- Materials
 - Set list of different materials/tools
- Build Challenges
 - Cleaning all the parts
 - Assembling the parts
 - Would likely need the help of Mr. Teifert
 - Tolerances
- Electronics
 - Not much completed yet for electronics
 - Need to get more detailed on what electronics, connections cables, and software for data acquisition



Preliminary Design Review

Spring 2024!



Timeline

Week 4-6

Complete design setup

Design Stage 1: Record and analyze results

Design Stage 2: Record and analyze results

Week 1-3

Finalize Design Changes

Order machined parts

New SIMION Simulation

LabView and Matlab Code

Figure out Electronics

Week 7-11

Design Stage 3: Record and analyze results

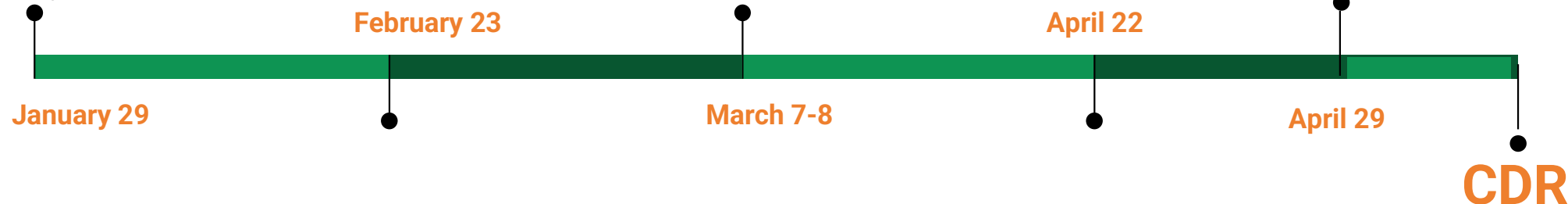
Design Stage 4: Record and analyze results

Write out Final results

Week 11-12

Buffer period:

Accounts for additional time needed to troubleshoot or delay in delivery of materials



Week 12: CDR and conclusion of Project!



Contingency Plan

Most likely to go wrong:

- Delay on parts being delivered
- Absence from sickness, break, etc.
- Troubleshoot design or re-evaluate design approach
- Issue with Labview/ analysis results

Respective Contingency

- Look at similar tools in lab or rethink design approach
- Buffer time: More time factored into each process and additional time set aside
- Communication is mandatory for all issues during weekly meetings. Troubleshooting will be done in teams responsible for each design stages/ software.



Project Management

- Task Assignment and Organization - Notion
 - Team members are responsible for updating Notion on their task title, task progress, questions, and/or concerns.
 - Each part of project (Design Setup, Software, Design Stages) will be assigned a section in Notion. Team members are expected to post data, additional information, and questions in their relevant sections.
- Weekly Reports
 - Team members are expected to present their status on project, demonstrate progress, and bring up questions/ concerns on Monday meetings.



Preliminary Design Review

Risk and Safety Concerns



Contamination and Additional Risk

- Contamination of Science Chamber
 - Internal parts/ cables not cleaned thoroughly
 - Trapped air within design
 - Unaccounted contaminants
- Additional Risks
 - High voltage can cause damage to electronics
 - Voltage discharge in Science Chamber
 - Any work done in Science Chamber can be a risk to other apparatus in the chamber
 - Risks while handling instruments; ensuring instruments are fully discharged when removing them from Science Chamber; general lab safety

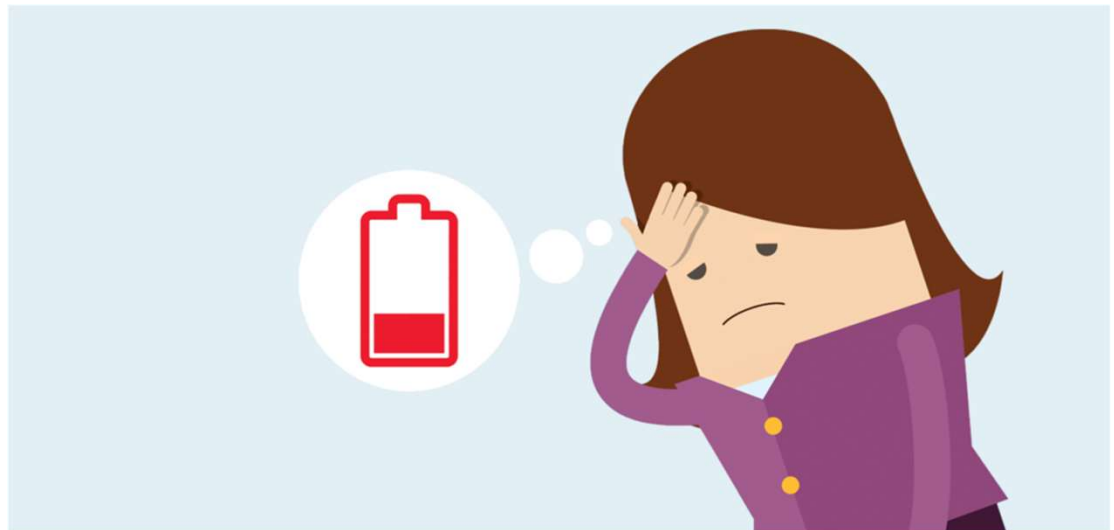


Minimize Risks

- Following a detailed written set of procedure/ guidelines
 - Ex. Placement of instrument inside Science Chamber, High Voltage Safety Usage
 - Work in teams of two or more
- Proper making/ handling of instruments
 - Ensure all cables are connected properly
 - Double checks by members of the team
- Run RGA scans, analyze results, and make changes/ proceed accordingly
- Appropriate safety gear
- Additional Concerns: Reach out to expertise
 - Prof. McComas, Dr. Rankin, Mr. Fennimore, Mr. Teifert, Post-docs

Safety Concerns

- High Voltage circuitry
- Possible injuries during assembly
- Handling of breakable lab instruments
- Exhaustion



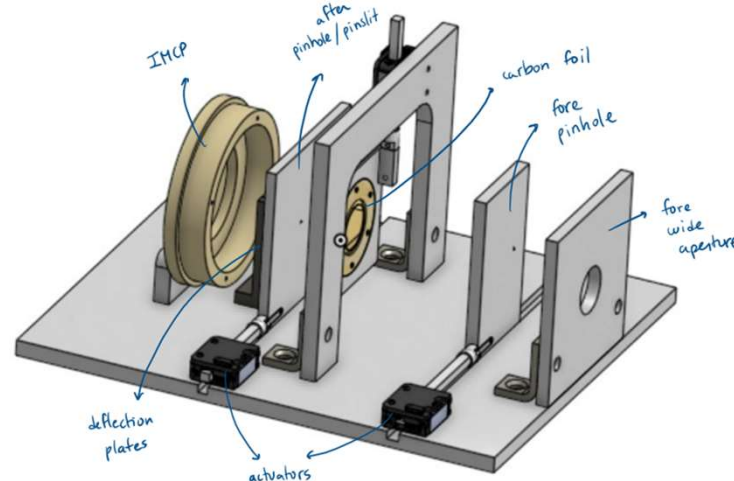


Preliminary Design Review

Summary

Summary

- **Goal:** Design of a simple apparatus that will help us learn more about ultra-thin carbon foils. Additional adjustments that can be added to learn other scientific results.
- **Design:**



- **Timeline:** Checkpoints made throughout the semester to ensure design setup and design stages are completed on time
- Risks, and safety concerns identified
- Suggestions?

Acknowledgements



Thank you everyone in Space Physics for teaching us about space flight instrumentation/ your research, demonstrating your day-to-day tasks, and helping/guiding us throughout our process!



Thank you!





Works Cited

- Allegrini, F., McComas, D. J., Young, D. T., Berthelier, J.-J., Covinhes, J., Illiano, J.-M., Riou, J.-F., Funsten, H. O., & Harper, R. W. (2006). Energy loss of 1–50keV H, He, C, N, O, Ne, and Ar ions transmitted through thin carbon foils. *Review of Scientific Instruments*, 77(4), 044501.
<https://doi.org/10.1063/1.2185490>
- Allegrini, F., Ebert, R. W., & Funsten, H. O. (2016). Carbon foils for space plasma instrumentation. *Journal of Geophysical Research: Space Physics*, 121(5), 3931–3950.
<https://doi.org/10.1002/2016JA022570>
- Allegrini, F., Ebert, R. W., Fuselier, S. A., Nicolaou, G., Bedworth, P., Sinton, S., & Trattner, K. J. (2014). Charge state of ~1 to 50 keV ions after passing through graphene and ultrathin carbon foils. *Optical Engineering*, 53(2), 024101. <https://doi.org/10.1117/1.OE.53.2.024101>
- Ebert, R. W., Allegrini, F., Fuselier, S. A., Nicolaou, G., Bedworth, P., Sinton, S., & Trattner, K. J. (2014). Angular scattering of 1–50 keV ions through graphene and thin carbon foils: Potential applications for space plasma instrumentation. *Review of Scientific Instruments*, 85(3), 033302.
<https://doi.org/10.1063/1.4866850>



Works Cited

- Funsten, H. O., Barraclough, B. L., & McComas, D. J. (1992). Pinhole detection in thin foils used in space plasma diagnostic instrumentation. *Review of Scientific Instruments*, 63(10), 4741–4743.
<https://doi.org/10.1063/1.1143626>
- Holeňák, R., Lohmann, S., Sekula, F., & Primetzhofer, D. (2021). Simultaneous assessment of energy, charge state and angular distribution for medium energy ions interacting with ultra-thin self-supporting targets: A time-of-flight approach. *Vacuum*, 185, 109988.
<https://doi.org/10.1016/j.vacuum.2020.109988>
- Meyer, L. (1971). Plural and multiple scattering of low-energy heavy particles in solids. *Physica Status Solidi (b)*, 44(1), 253–268. <https://doi.org/10.1002/pssb.2220440127>
- Ritzau, S. M., & Baragiola, R. A. (1998). Electron emission from carbon foils induced by keV ions. *Physical Review B*, 58(5), 2529–2538. <https://doi.org/10.1103/PhysRevB.58.2529>
- Vira, A. D., Fernandes, P. A., Funsten, H. O., Morley, S. K., Yamaguchi, H., Liu, F., & Moody, N. A. (2020). Angular scattering of protons through ultrathin graphene foils: Application for time-of-flight instrumentation. *Review of Scientific Instruments*, 91(3), 033302.
<https://doi.org/10.1063/1.5134768>

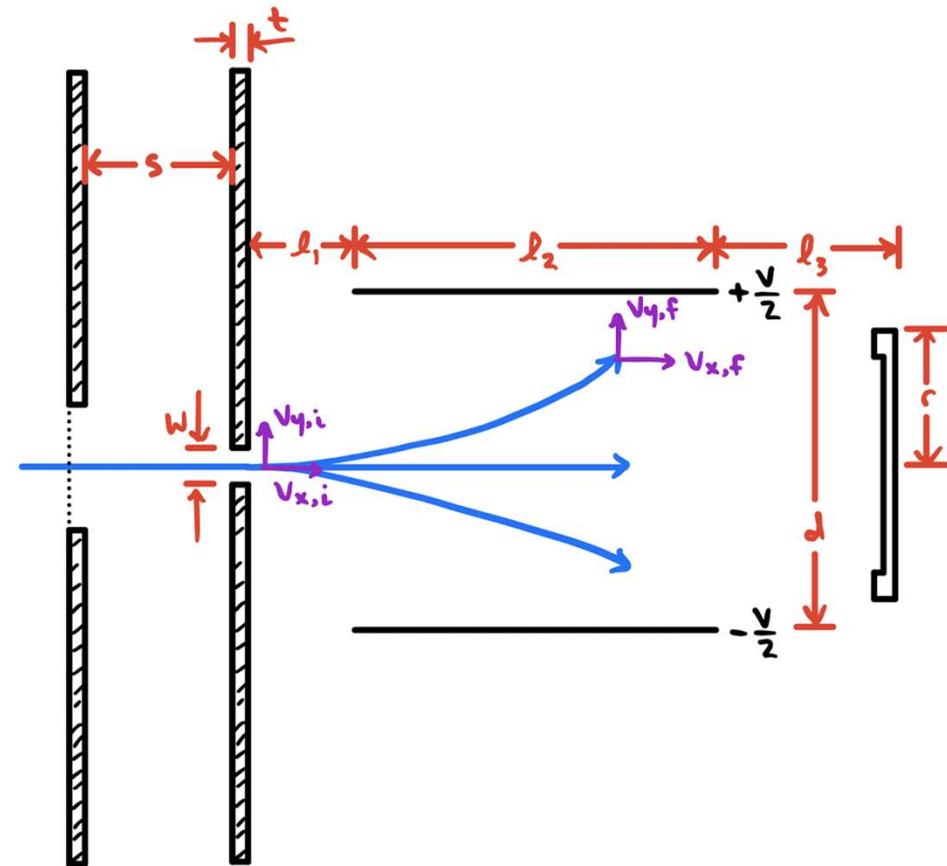


Preliminary Design Review

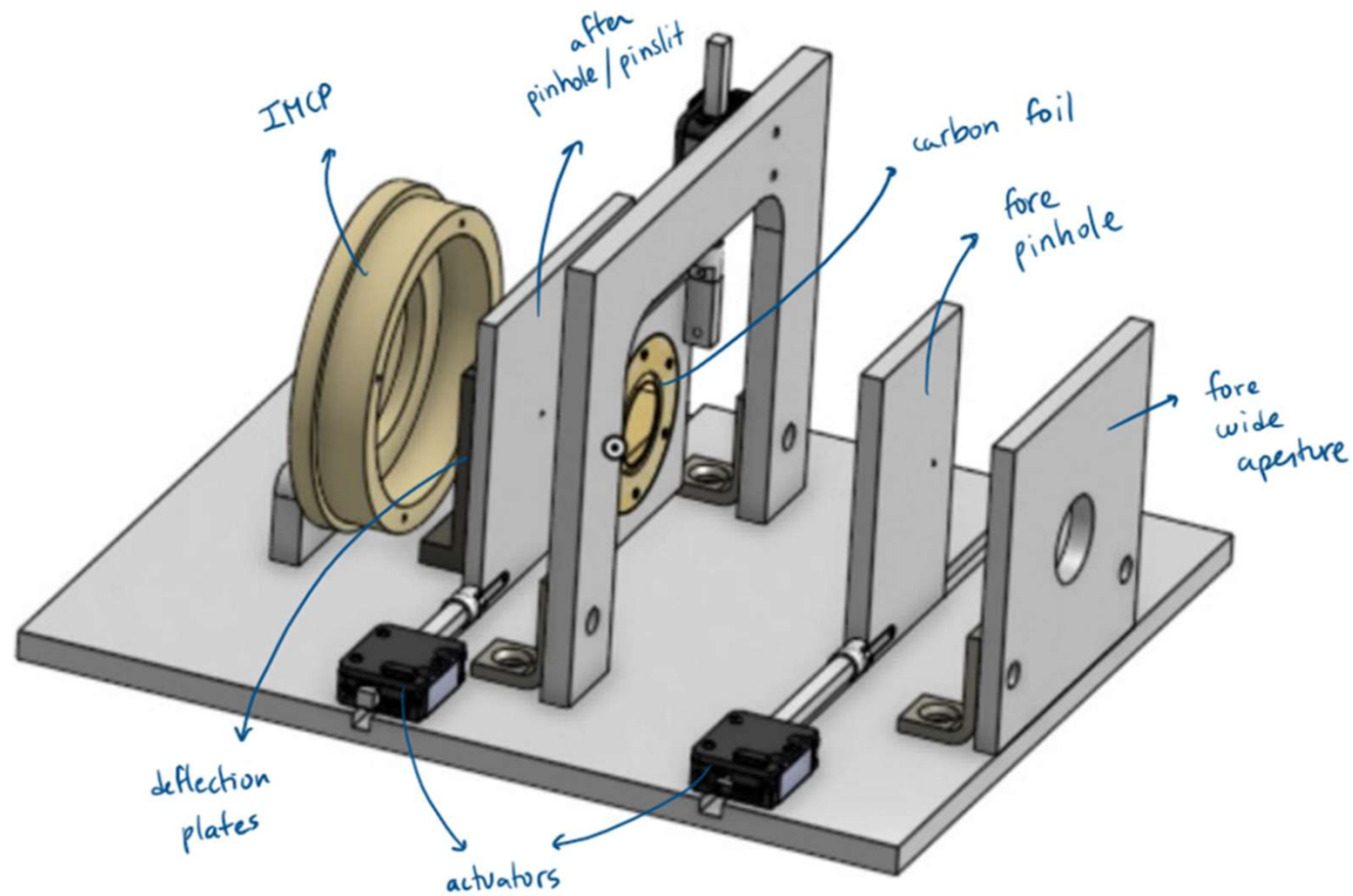
Backup

Design Specifications: Possible Parameters

Parameter	Value
E	650 – 1250 eV
V	700 – 2500 V
q	$(-1e) - (+2e)$
r	20 mm
s	20 mm
d	12 mm
w	1 mm
t	4 mm
ℓ_1	2 mm
ℓ_2	8 mm
ℓ_3	12 mm



Instrument Assembly



CAD design

Instrument Assembly

